

# Final2

January 5, 2025

## 1 DATA ANALYSIS

```
[468]: # import python libraries

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt # visualizing data
%matplotlib inline
import seaborn as sns
```

```
[470]: # import a dataset file

data = pd.read_csv('retail_store_inventory.csv', encoding= 'unicode_escape',
↳index_col=False)
```

2 Exploratory Data Analysis (EDA) and statistical analysis involve various methods and techniques to understand, explore, and prepare data before cleaning and processing.

### 3 Methods Included in EDA, Statistical Analysis, and Data Exploration

- Exploratory Data Analysis (EDA) involves systematically examining the data to uncover patterns, identify anomalies, test hypotheses, and check assumptions.
- Focusing on exploration before cleaning and processing, checking outliers, and statistical analysis.

#### 4 1. Understand the Dataset

- **Data Description:** Understand the dataset's context, source, and domain.
- **Metadata Exploration:** Review the data dictionary, column descriptions, and units of measurement.

## 5 a. Context, Source, and Domain

- **Context:** This **dataset** provides insights into retail store inventory, focusing on sales, demand forecasting, and external influencing factors. It can be used to analyze **stock levels**, **sales trends**, and **the impact of promotions or seasonal variations**.
- **Source:** Determine if the **dataset** is real-world (e.g., company-provided) or synthetic. If it's real-world, ensure the data complies with privacy and data usage regulations (e.g., GDPR).
- **Domain:** Retail and Supply Chain Management, specifically Inventory Optimization and Sales Forecasting.

[472]: data

```
[472]:      Date Store ID Product ID      Category Region  Inventory Level \
0      2022-01-01      S001      P0001      Groceries North      231.0
1      2022-01-01      S001      P0002              Toys  South      204.0
2      2022-01-01      S001      P0003              Toys  West      102.0
3      2022-01-01      S001      P0004              Toys  North      469.0
4      2022-01-01      S001      P0005 Electronics  East      166.0
...      ...      ...      ...      ...      ...      ...
49520    2023-05-11      S002      P0001      Clothing  West      385.0
49521    2023-05-11      S002      P0002              Toys  South      443.0
49522    2023-05-11      S002      P0003      Groceries North      493.0
49523    2023-05-11      S002      P0004      Furniture  West      498.0
49524    2023-05-11      S002      P000      NaN      NaN      NaN
```

```
      Units Sold  Units Ordered  Demand Forecast  Price  Discount \
0      127.0      55.0      135.47  33.50      20.0
1      150.0      66.0      144.04  63.01      20.0
2      65.0      51.0      74.02  27.99      10.0
3      61.0      164.0      62.18  32.72      10.0
4      14.0      135.0      9.26  73.64      0.0
...      ...      ...      ...      ...      ...
49520      143.0      36.0      143.83  42.12      15.0
49521      77.0      53.0      79.59  33.25      20.0
49522      100.0      23.0      94.13  21.28      10.0
49523      461.0      197.0      454.53  99.63      20.0
49524      NaN      NaN      NaN      NaN      NaN
```

```
      Weather Condition  Holiday/Promotion  Competitor Pricing Seasonality
0      Rainy      0.0      29.69      Autumn
1      Sunny      0.0      66.16      Autumn
2      Sunny      1.0      31.32      Summer
3      Cloudy      1.0      34.74      Autumn
4      Sunny      0.0      68.95      Summer
...      ...      ...      ...      ...
49520      Sunny      0.0      47.07      Spring
49521      Cloudy      0.0      32.05      Winter
```

|       |       |     |        |        |
|-------|-------|-----|--------|--------|
| 49522 | Snowy | 1.0 | 19.78  | Spring |
| 49523 | Snowy | 0.0 | 103.17 | Winter |
| 49524 | NaN   | NaN | NaN    | NaN    |

[49525 rows x 15 columns]

```
[474]: pd.Series(data['Category'])
```

```
[474]: 0      Groceries
1      Toys
2      Toys
3      Toys
4      Electronics
...
49520   Clothing
49521     Toys
49522   Groceries
49523   Furniture
49524     NaN
Name: Category, Length: 49525, dtype: object
```

```
[476]: ax = data.sort_values("Category", axis=0, inplace=False, ascending=True)
ax
ac = ax.reset_index()
ac
```

```
[476]:
```

|       | index | Date       | Store ID | Product ID | Category | Region \ |
|-------|-------|------------|----------|------------|----------|----------|
| 0     | 16450 | 2022-06-14 | S003     | P0011      | Clothing | West     |
| 1     | 35635 | 2022-12-23 | S002     | P0016      | Clothing | West     |
| 2     | 12161 | 2022-05-02 | S004     | P0002      | Clothing | South    |
| 3     | 12162 | 2022-05-02 | S004     | P0003      | Clothing | South    |
| 4     | 27454 | 2022-10-02 | S003     | P0015      | Clothing | South    |
| ...   | ...   | ...        | ...      | ...        | ...      | ...      |
| 49520 | 39797 | 2023-02-02 | S005     | P0018      | Toys     | East     |
| 49521 | 19144 | 2022-07-11 | S003     | P0005      | Toys     | East     |
| 49522 | 19260 | 2022-07-12 | S004     | P0001      | Toys     | East     |
| 49523 | 24761 | 2022-09-05 | S004     | P0002      | Toys     | South    |
| 49524 | 49524 | 2023-05-11 | S002     | P000       | NaN      | NaN      |

|       | Inventory Level | Units Sold | Units Ordered | Demand Forecast | Price \ |
|-------|-----------------|------------|---------------|-----------------|---------|
| 0     | 407.0           | 236.0      | 188.0         | 241.05          | 42.53   |
| 1     | 380.0           | 156.0      | 106.0         | 174.74          | 85.81   |
| 2     | 56.0            | 2.0        | 137.0         | 7.53            | 93.03   |
| 3     | 227.0           | 186.0      | 128.0         | 191.07          | 17.54   |
| 4     | 261.0           | 27.0       | 89.0          | 39.78           | 47.43   |
| ...   | ...             | ...        | ...           | ...             | ...     |
| 49520 | 466.0           | 138.0      | 111.0         | 142.03          | 73.70   |

|       |       |       |       |        |       |
|-------|-------|-------|-------|--------|-------|
| 49521 | 445.0 | 184.0 | 170.0 | 200.12 | 56.95 |
| 49522 | 382.0 | 259.0 | 29.0  | 255.68 | 39.99 |
| 49523 | 144.0 | 50.0  | 37.0  | 57.78  | 25.39 |
| 49524 | NaN   | NaN   | NaN   | NaN    | NaN   |

|       | Discount | Weather Condition | Holiday/Promotion | Competitor Pricing \ |
|-------|----------|-------------------|-------------------|----------------------|
| 0     | 20.0     | Rainy             | 1.0               | 46.37                |
| 1     | 10.0     | Snowy             | 1.0               | 85.47                |
| 2     | 15.0     | Snowy             | 0.0               | 94.60                |
| 3     | 10.0     | Rainy             | 1.0               | 15.63                |
| 4     | 20.0     | Snowy             | 1.0               | 42.59                |
| ...   | ...      | ...               | ...               | ...                  |
| 49520 | 0.0      | Cloudy            | 1.0               | 74.42                |
| 49521 | 20.0     | Rainy             | 1.0               | 55.05                |
| 49522 | 15.0     | Cloudy            | 1.0               | 35.69                |
| 49523 | 10.0     | Snowy             | 1.0               | 29.34                |
| 49524 | NaN      | NaN               | NaN               | NaN                  |

|       | Seasonality |
|-------|-------------|
| 0     | Winter      |
| 1     | Winter      |
| 2     | Autumn      |
| 3     | Winter      |
| 4     | Autumn      |
| ...   | ...         |
| 49520 | Summer      |
| 49521 | Winter      |
| 49522 | Summer      |
| 49523 | Autumn      |
| 49524 | NaN         |

[49525 rows x 16 columns]

```
[478]: ac.drop(columns='index')
```

```
[478]:
```

|       | Date       | Store ID | Product ID | Category | Region | Inventory Level \ |
|-------|------------|----------|------------|----------|--------|-------------------|
| 0     | 2022-06-14 | S003     | P0011      | Clothing | West   | 407.0             |
| 1     | 2022-12-23 | S002     | P0016      | Clothing | West   | 380.0             |
| 2     | 2022-05-02 | S004     | P0002      | Clothing | South  | 56.0              |
| 3     | 2022-05-02 | S004     | P0003      | Clothing | South  | 227.0             |
| 4     | 2022-10-02 | S003     | P0015      | Clothing | South  | 261.0             |
| ...   | ...        | ...      | ...        | ...      | ...    | ...               |
| 49520 | 2023-02-02 | S005     | P0018      | Toys     | East   | 466.0             |
| 49521 | 2022-07-11 | S003     | P0005      | Toys     | East   | 445.0             |
| 49522 | 2022-07-12 | S004     | P0001      | Toys     | East   | 382.0             |
| 49523 | 2022-09-05 | S004     | P0002      | Toys     | South  | 144.0             |
| 49524 | 2023-05-11 | S002     | P000       | NaN      | NaN    | NaN               |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 0     | 236.0      | 188.0         | 241.05          | 42.53 | 20.0     |   |
| 1     | 156.0      | 106.0         | 174.74          | 85.81 | 10.0     |   |
| 2     | 2.0        | 137.0         | 7.53            | 93.03 | 15.0     |   |
| 3     | 186.0      | 128.0         | 191.07          | 17.54 | 10.0     |   |
| 4     | 27.0       | 89.0          | 39.78           | 47.43 | 20.0     |   |
| ...   | ...        | ...           | ...             | ...   | ...      |   |
| 49520 | 138.0      | 111.0         | 142.03          | 73.70 | 0.0      |   |
| 49521 | 184.0      | 170.0         | 200.12          | 56.95 | 20.0     |   |
| 49522 | 259.0      | 29.0          | 255.68          | 39.99 | 15.0     |   |
| 49523 | 50.0       | 37.0          | 57.78           | 25.39 | 10.0     |   |
| 49524 | NaN        | NaN           | NaN             | NaN   | NaN      |   |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality |
|-------|-------------------|-------------------|--------------------|-------------|
| 0     | Rainy             | 1.0               | 46.37              | Winter      |
| 1     | Snowy             | 1.0               | 85.47              | Winter      |
| 2     | Snowy             | 0.0               | 94.60              | Autumn      |
| 3     | Rainy             | 1.0               | 15.63              | Winter      |
| 4     | Snowy             | 1.0               | 42.59              | Autumn      |
| ...   | ...               | ...               | ...                | ...         |
| 49520 | Cloudy            | 1.0               | 74.42              | Summer      |
| 49521 | Rainy             | 1.0               | 55.05              | Winter      |
| 49522 | Cloudy            | 1.0               | 35.69              | Summer      |
| 49523 | Snowy             | 1.0               | 29.34              | Autumn      |
| 49524 | NaN               | NaN               | NaN                | NaN         |

[49525 rows x 15 columns]

## 6 Column Name, Description, Type, Units/Format

- **Date** “The date of the record, useful for time-series analysis.” `DateTime YYYY-MM-DD`
- **Store ID** “Unique identifier for each retail store.” `Categorical Alphanumeric` or `Numeric`
- **Product ID** “Unique identifier for each product.” `Categorical Alphanumeric` or `Numeric`
- **Category** “Product category (e.g., Electronics, Groceries).” `Categorical Text`
- **Region** “Geographical region of the store.” `Categorical Text`
- **Inventory Level** “Number of units currently available in stock.” `Numeric Integer`
- **Units Sold** “Number of units sold on the given date.” `Numeric Integer`
- **Units Ordered** “Number of units ordered by the store for replenishment.” `Numeric Integer`
- **Demand Forecast** “Predicted number of units expected to be sold.” `Numeric Integer`
- **Price** “Selling price per unit of the product.” `Numeric Currency` (e.g., USD)

- **Discount** “Discount applied to the product price, either in percentage or absolute value.”  
Numeric Percentage or Currency
- **Seasonality** “Indicates the seasonal demand pattern (e.g., high, medium, low).”  
Categorical Text
- **Weather Condition** “Weather on the date (e.g., Sunny, Rainy, Snowy).” Categorical Text
- **Holiday/Promotion** “Indicates whether the date had a holiday or store promotion.”  
Categorical Yes/No
- **Competitor Pricing** “Price of the same or similar product at a competitor’s store.” Numeric  
Currency (e.g., USD)

## 7 b. Initial Observations

- **Key Time-Series Columns:** Date, Seasonality, Weather Condition.
- **Causal Factors:** Holiday/Promotion, Competitor Pricing, Weather Condition.
- **Sales-Driven Metrics:** Units Sold, Price, Discount.
- **Operational Metrics:** Inventory Level, Units Ordered, Demand Forecast.
- **Categorical Dimensions:** Store ID, Product ID, Category, Region.

[ ]:

## 8 2. Load and Inspect the Data

- **Head and Tail:** Use `data.head()` and `data.tail()` to view the first and last rows of the dataset.
- **Shape:** `data.shape` provides the number of rows and columns.
- **Data Types:** Check data types with `data.info()`.
- **Missing Values:** Use `data.isnull().sum()` to identify missing values.
- **Duplicate Rows:** Check for duplicates using `data.duplicated().sum()`.

[480]: `data.head()` *#for first rows of the dataset by default show 5*

```
[480]:
```

|   | Date       | Store ID | Product ID | Category    | Region | Inventory Level | \ |
|---|------------|----------|------------|-------------|--------|-----------------|---|
| 0 | 2022-01-01 | S001     | P0001      | Groceries   | North  | 231.0           |   |
| 1 | 2022-01-01 | S001     | P0002      | Toys        | South  | 204.0           |   |
| 2 | 2022-01-01 | S001     | P0003      | Toys        | West   | 102.0           |   |
| 3 | 2022-01-01 | S001     | P0004      | Toys        | North  | 469.0           |   |
| 4 | 2022-01-01 | S001     | P0005      | Electronics | East   | 166.0           |   |

|   | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|---|------------|---------------|-----------------|-------|----------|---|
| 0 | 127.0      | 55.0          | 135.47          | 33.50 | 20.0     |   |
| 1 | 150.0      | 66.0          | 144.04          | 63.01 | 20.0     |   |

|   |      |       |       |       |      |
|---|------|-------|-------|-------|------|
| 2 | 65.0 | 51.0  | 74.02 | 27.99 | 10.0 |
| 3 | 61.0 | 164.0 | 62.18 | 32.72 | 10.0 |
| 4 | 14.0 | 135.0 | 9.26  | 73.64 | 0.0  |

|   | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality |
|---|-------------------|-------------------|--------------------|-------------|
| 0 | Rainy             | 0.0               | 29.69              | Autumn      |
| 1 | Sunny             | 0.0               | 66.16              | Autumn      |
| 2 | Sunny             | 1.0               | 31.32              | Summer      |
| 3 | Cloudy            | 1.0               | 34.74              | Autumn      |
| 4 | Sunny             | 0.0               | 68.95              | Summer      |

```
[482]: data.tail()      ##for last rows of the dataset by default show 5
```

```
[482]:      Date Store ID Product ID  Category Region  Inventory Level \
49520  2023-05-11   S002    P0001  Clothing   West        385.0
49521  2023-05-11   S002    P0002    Toys    South        443.0
49522  2023-05-11   S002    P0003  Groceries  North        493.0
49523  2023-05-11   S002    P0004  Furniture  West        498.0
49524  2023-05-11   S002    P000    NaN     NaN          NaN
```

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 49520 | 143.0      | 36.0          | 143.83          | 42.12 | 15.0     |   |
| 49521 | 77.0       | 53.0          | 79.59           | 33.25 | 20.0     |   |
| 49522 | 100.0      | 23.0          | 94.13           | 21.28 | 10.0     |   |
| 49523 | 461.0      | 197.0         | 454.53          | 99.63 | 20.0     |   |
| 49524 | NaN        | NaN           | NaN             | NaN   | NaN      |   |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality |
|-------|-------------------|-------------------|--------------------|-------------|
| 49520 | Sunny             | 0.0               | 47.07              | Spring      |
| 49521 | Cloudy            | 0.0               | 32.05              | Winter      |
| 49522 | Snowy             | 1.0               | 19.78              | Spring      |
| 49523 | Snowy             | 0.0               | 103.17             | Winter      |
| 49524 | NaN               | NaN               | NaN                | NaN         |

```
[484]: data.shape      #for shape of a dataset
```

```
[484]: (49525, 15)
```

```
[486]: data.dtypes      #for check data dtypes
```

```
[486]: Date           object
Store ID           object
Product ID         object
Category           object
Region            object
Inventory Level     float64
Units Sold         float64
```

```

Units Ordered      float64
Demand Forecast    float64
Price              float64
Discount           float64
Weather Condition  object
Holiday/Promotion  float64
Competitor Pricing float64
Seasonality        object
dtype: object

```

```
[488]: data.info()      #for check data type
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49525 entries, 0 to 49524
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Date                  49525 non-null  object
 1   Store ID              49525 non-null  object
 2   Product ID            49525 non-null  object
 3   Category              49524 non-null  object
 4   Region                49524 non-null  object
 5   Inventory Level       49524 non-null  float64
 6   Units Sold            49524 non-null  float64
 7   Units Ordered         49524 non-null  float64
 8   Demand Forecast       49524 non-null  float64
 9   Price                 49524 non-null  float64
10  Discount              49524 non-null  float64
11  Weather Condition     49524 non-null  object
12  Holiday/Promotion     49524 non-null  float64
13  Competitor Pricing    49524 non-null  float64
14  Seasonality           49524 non-null  object
dtypes: float64(8), object(7)
memory usage: 5.7+ MB

```

```
[490]: data.isnull().sum()      #for check missing values
```

```

[490]: Date                0
Store ID                  0
Product ID                0
Category                  1
Region                    1
Inventory Level           1
Units Sold                1
Units Ordered             1
Demand Forecast           1
Price                     1

```

```
Discount          1
Weather Condition  1
Holiday/Promotion  1
Competitor Pricing 1
Seasonality       1
dtype: int64
```

```
[492]: data.duplicated().sum()           #for check duplication row-wise
```

```
[492]: 0
```

## 9 3. Handling missing\_\_values(NaN), Cleaning a Dataset and Explore

```
[495]: data['Category'].fillna(data['Category'].mode()[0], inplace=True)
```

```
[497]: data['Region'].fillna(data['Region'].mode()[0], inplace=True)
```

```
[499]: data['Inventory Level'].fillna(data['Inventory Level'].mean(), inplace=True)
```

```
[501]: data['Units Sold'].fillna(data['Units Sold'].mean(), inplace=True)
```

```
[503]: data['Units Ordered'].fillna(data['Units Ordered'].mean(), inplace=True)
```

```
[505]: data['Demand Forecast'].fillna(data['Demand Forecast'].median(), inplace=True)
```

```
[507]: data['Price'] = data['Price'].fillna(data['Price'].median())
```

```
[509]: data['Discount'] = data['Discount'].fillna(data['Discount'].median())
```

```
[511]: data['Weather Condition'] = data['Weather Condition'].fillna(data['Weather_
↪Condition'].mode()[0])
```

```
[513]: data['Seasonality'].fillna(data['Seasonality'].mode()[0], inplace=True)
```

```
[515]: data['Holiday/Promotion'].fillna(data['Holiday/Promotion'].mean(), inplace=True)
```

```
[517]: data['Competitor Pricing'].fillna(method='ffill', inplace=True)
```

## 10 Changing data Types of the columns

### 11 Method 1: Using astype() Method

The `astype()` method allows you to change the data type of a column.

```
[520]: data['Inventory Level'] = data['Inventory Level'].astype(int)
```

## 12 Method 2: Using apply() Method with int()

If we want more control or need to apply custom logic, use `apply()` with a lambda function.

```
[523]: data['Units Sold'] = data['Units Sold'].apply(lambda x: int(x))
```

## 13 Method 3: Using numpy for Multiple Columns

If you need to convert multiple columns, you can use `numpy`.

```
[526]: data[['Units Ordered', 'Demand Forecast', 'Discount']] = data[['Units Ordered', 'Demand Forecast', 'Discount']].astype(np.int64)
```

- Converting float to integer will truncate the decimal part.
- If rounding is required before conversion, use the `round()` method.

```
[529]: data[['Price', 'Competitor Pricing', 'Holiday/Promotion']] = data[['Price', 'Competitor Pricing', 'Holiday/Promotion']].round().astype(int)
```

```
[531]: ### Explore dataset after handling nan values
data.head()
```

```
[531]:      Date Store ID Product ID      Category Region  Inventory Level \
0  2022-01-01   S001    P0001    Groceries  North         231
1  2022-01-01   S001    P0002         Toys  South         204
2  2022-01-01   S001    P0003         Toys  West          102
3  2022-01-01   S001    P0004         Toys  North         469
4  2022-01-01   S001    P0005  Electronics  East          166
```

```
      Units Sold  Units Ordered  Demand Forecast  Price  Discount \
0          127           55          135      34         20
1          150           66          144      63         20
2           65           51           74      28         10
3           61          164           62      33         10
4           14          135           9       74          0
```

```
      Weather Condition  Holiday/Promotion  Competitor Pricing  Seasonality
0          Rainy          0          30      Autumn
1          Sunny          0          66      Autumn
2          Sunny          1          31      Summer
3          Cloudy          1          35      Autumn
4          Sunny          0          69      Summer
```

```
[533]: ##Explore dataset after cleaning nan values
data.tail()
```

```
[533]:
```

|       | Date       | Store ID | Product ID | Category  | Region | Inventory Level | \ |
|-------|------------|----------|------------|-----------|--------|-----------------|---|
| 49520 | 2023-05-11 | S002     | P0001      | Clothing  | West   | 385             |   |
| 49521 | 2023-05-11 | S002     | P0002      | Toys      | South  | 443             |   |
| 49522 | 2023-05-11 | S002     | P0003      | Groceries | North  | 493             |   |
| 49523 | 2023-05-11 | S002     | P0004      | Furniture | West   | 498             |   |
| 49524 | 2023-05-11 | S002     | P000       | Groceries | West   | 274             |   |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 49520 | 143        | 36            | 143             | 42    | 15       |   |
| 49521 | 77         | 53            | 79              | 33    | 20       |   |
| 49522 | 100        | 23            | 94              | 21    | 10       |   |
| 49523 | 461        | 197           | 454             | 100   | 20       |   |
| 49524 | 136        | 110           | 112             | 55    | 10       |   |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality |
|-------|-------------------|-------------------|--------------------|-------------|
| 49520 | Sunny             | 0                 | 47                 | Spring      |
| 49521 | Cloudy            | 0                 | 32                 | Winter      |
| 49522 | Snowy             | 1                 | 20                 | Spring      |
| 49523 | Snowy             | 0                 | 103                | Winter      |
| 49524 | Snowy             | 1                 | 103                | Spring      |

## 14 Convert the data\_type of Date col in Datetime Series.

- Add a column of Year, Month, and Month Name in Aplhabetical format.

```
[536]: data['Date'] = pd.to_datetime(data['Date'])
```

```
[538]: data['Year'] = data['Date'].dt.year
```

```
[540]: data['Month'] = data['Date'].dt.month
```

```
[542]: data[['Year', 'Month']] = data[['Year', 'Month']].astype(np.int64)
data.dtypes      ## Convert dtypes
```

```
[542]: Date          datetime64[ns]
Store ID           object
Product ID         object
Category           object
Region            object
Inventory Level     int64
Units Sold         int64
Units Ordered      int64
Demand Forecast    int64
Price             int64
```

```
Discount          int64
Weather Condition  object
Holiday/Promotion  int64
Competitor Pricing int64
Seasonality        object
Year              int64
Month             int64
dtype: object
```

```
[544]: ## Add a month name column in a dataset
data['Name of Month'] = data['Date'].dt.strftime('%b')
```

```
[546]: ## After all cleaning, handling values or add additional columns
print(f"Rows: {data.shape[0]}, Columns: {data.shape[1]}")
```

Rows: 49525, Columns: 18

```
[549]: data.head()
```

```
[549]:
```

|   | Date       | Store ID | Product ID | Category    | Region | Inventory | Level \ |
|---|------------|----------|------------|-------------|--------|-----------|---------|
| 0 | 2022-01-01 | S001     | P0001      | Groceries   | North  | 231       |         |
| 1 | 2022-01-01 | S001     | P0002      | Toys        | South  | 204       |         |
| 2 | 2022-01-01 | S001     | P0003      | Toys        | West   | 102       |         |
| 3 | 2022-01-01 | S001     | P0004      | Toys        | North  | 469       |         |
| 4 | 2022-01-01 | S001     | P0005      | Electronics | East   | 166       |         |

|   | Units Sold | Units Ordered | Demand Forecast | Price | Discount \ |
|---|------------|---------------|-----------------|-------|------------|
| 0 | 127        | 55            | 135             | 34    | 20         |
| 1 | 150        | 66            | 144             | 63    | 20         |
| 2 | 65         | 51            | 74              | 28    | 10         |
| 3 | 61         | 164           | 62              | 33    | 10         |
| 4 | 14         | 135           | 9               | 74    | 0          |

|   | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality | Year \ |
|---|-------------------|-------------------|--------------------|-------------|--------|
| 0 | Rainy             | 0                 | 30                 | Autumn      | 2022   |
| 1 | Sunny             | 0                 | 66                 | Autumn      | 2022   |
| 2 | Sunny             | 1                 | 31                 | Summer      | 2022   |
| 3 | Cloudy            | 1                 | 35                 | Autumn      | 2022   |
| 4 | Sunny             | 0                 | 69                 | Summer      | 2022   |

|   | Month | Name of Month |
|---|-------|---------------|
| 0 | 1     | Jan           |
| 1 | 1     | Jan           |
| 2 | 1     | Jan           |
| 3 | 1     | Jan           |
| 4 | 1     | Jan           |

## 15 4. Statistical Summary

- **Summary Statistics:** Use `data.describe()` for numerical columns and `data.describe(include='object')` for categorical columns.
- **Central Tendency and Dispersion:** Analyze mean, median, mode, standard deviation, variance, and range.

```
[551]: data.describe() #use for numerically columns or data
```

```
[551]:
```

|       | Date                          | Inventory Level | Units Sold   | \ |
|-------|-------------------------------|-----------------|--------------|---|
| count | 49525                         | 49525.000000    | 49525.000000 |   |
| mean  | 2022-09-05 03:00:16.355375872 | 274.636729      | 136.389258   |   |
| min   | 2022-01-01 00:00:00           | 50.000000       | 0.000000     |   |
| 25%   | 2022-05-04 00:00:00           | 163.000000      | 48.000000    |   |
| 50%   | 2022-09-05 00:00:00           | 274.000000      | 108.000000   |   |
| 75%   | 2023-01-07 00:00:00           | 387.000000      | 202.000000   |   |
| max   | 2023-05-11 00:00:00           | 500.000000      | 496.000000   |   |
| std   | NaN                           | 129.727003      | 108.938973   |   |

|       | Units Ordered | Demand Forecast | Price        | Discount     | \ |
|-------|---------------|-----------------|--------------|--------------|---|
| count | 49525.000000  | 49525.000000    | 49525.000000 | 49525.000000 |   |
| mean  | 110.046118    | 140.931045      | 55.332458    | 10.021807    |   |
| min   | 20.000000     | -9.000000       | 10.000000    | 0.000000     |   |
| 25%   | 65.000000     | 53.000000       | 33.000000    | 5.000000     |   |
| 50%   | 110.000000    | 112.000000      | 55.000000    | 10.000000    |   |
| 75%   | 155.000000    | 207.000000      | 78.000000    | 15.000000    |   |
| max   | 200.000000    | 512.000000      | 100.000000   | 20.000000    |   |
| std   | 52.333915     | 109.290107      | 26.038579    | 7.077242     |   |

|       | Holiday/Promotion | Competitor Pricing | Year         | Month        |
|-------|-------------------|--------------------|--------------|--------------|
| count | 49525.000000      | 49525.000000       | 49525.000000 | 49525.000000 |
| mean  | 0.500050          | 55.347542          | 2022.262998  | 5.518930     |
| min   | 0.000000          | 5.000000           | 2022.000000  | 1.000000     |
| 25%   | 0.000000          | 33.000000          | 2022.000000  | 3.000000     |
| 50%   | 1.000000          | 55.000000          | 2022.000000  | 5.000000     |
| 75%   | 1.000000          | 78.000000          | 2023.000000  | 8.000000     |
| max   | 1.000000          | 105.000000         | 2023.000000  | 12.000000    |
| std   | 0.500005          | 26.201560          | 0.440266     | 3.468399     |

```
[553]: data.describe(include='object') #use for categorical/textual_based_  

↪ columns
```

```
[553]:
```

|        | Store ID | Product ID | Category  | Region | Weather | Condition | Seasonality | \ |
|--------|----------|------------|-----------|--------|---------|-----------|-------------|---|
| count  | 49525    | 49525      | 49525     | 49525  |         | 49525     | 49525       |   |
| unique | 5        | 21         | 5         | 4      |         | 4         | 4           |   |
| top    | S001     | P0001      | Groceries | West   |         | Snowy     | Spring      |   |
| freq   | 9920     | 2477       | 10042     | 12412  |         | 12408     | 12443       |   |

|        | Name of Month |
|--------|---------------|
| count  | 49525         |
| unique | 12            |
| top    | Jan           |
| freq   | 6200          |

```
[555]: (data.notnull().sum()/len(data))* 100
```

```
[555]: Date          100.0
Store ID         100.0
Product ID       100.0
Category         100.0
Region          100.0
Inventory Level  100.0
Units Sold       100.0
Units Ordered    100.0
Demand Forecast  100.0
Price            100.0
Discount         100.0
Weather Condition 100.0
Holiday/Promotion 100.0
Competitor Pricing 100.0
Seasonality      100.0
Year             100.0
Month            100.0
Name of Month    100.0
dtype: float64
```

```
[557]: data_num = data.select_dtypes(include=['number'])
data_num
```

```
[557]:
```

|       | Inventory Level | Units Sold | Units Ordered | Demand Forecast | Price | \ |
|-------|-----------------|------------|---------------|-----------------|-------|---|
| 0     | 231             | 127        | 55            | 135             | 34    |   |
| 1     | 204             | 150        | 66            | 144             | 63    |   |
| 2     | 102             | 65         | 51            | 74              | 28    |   |
| 3     | 469             | 61         | 164           | 62              | 33    |   |
| 4     | 166             | 14         | 135           | 9               | 74    |   |
| ...   | ...             | ...        | ...           | ...             | ...   |   |
| 49520 | 385             | 143        | 36            | 143             | 42    |   |
| 49521 | 443             | 77         | 53            | 79              | 33    |   |
| 49522 | 493             | 100        | 23            | 94              | 21    |   |
| 49523 | 498             | 461        | 197           | 454             | 100   |   |
| 49524 | 274             | 136        | 110           | 112             | 55    |   |

|   | Discount | Holiday/Promotion | Competitor Pricing | Year | Month |
|---|----------|-------------------|--------------------|------|-------|
| 0 | 20       | 0                 | 30                 | 2022 | 1     |

|       |     |     |     |      |     |
|-------|-----|-----|-----|------|-----|
| 1     | 20  | 0   | 66  | 2022 | 1   |
| 2     | 10  | 1   | 31  | 2022 | 1   |
| 3     | 10  | 1   | 35  | 2022 | 1   |
| 4     | 0   | 0   | 69  | 2022 | 1   |
| ...   | ... | ... | ... | ...  | ... |
| 49520 | 15  | 0   | 47  | 2023 | 5   |
| 49521 | 20  | 0   | 32  | 2023 | 5   |
| 49522 | 10  | 1   | 20  | 2023 | 5   |
| 49523 | 20  | 0   | 103 | 2023 | 5   |
| 49524 | 10  | 1   | 103 | 2023 | 5   |

[49525 rows x 10 columns]

```
[559]: data_num.sum()
```

```
[559]: Inventory Level      13601384
Units Sold                6754678
Units Ordered             5450034
Demand Forecast           6979610
Price                    2740340
Discount                  496330
Holiday/Promotion         24765
Competitor Pricing        2741087
Year                     100152575
Month                    273325
dtype: int64
```

```
[561]: data_num.mean()
```

```
[561]: Inventory Level      274.636729
Units Sold                136.389258
Units Ordered             110.046118
Demand Forecast           140.931045
Price                     55.332458
Discount                  10.021807
Holiday/Promotion         0.500050
Competitor Pricing        55.347542
Year                     2022.262998
Month                     5.518930
dtype: float64
```

```
[563]: data_num.median()
```

```
[563]: Inventory Level      274.0
Units Sold                108.0
Units Ordered             110.0
Demand Forecast           112.0
```

|                    |        |
|--------------------|--------|
| Price              | 55.0   |
| Discount           | 10.0   |
| Holiday/Promotion  | 1.0    |
| Competitor Pricing | 55.0   |
| Year               | 2022.0 |
| Month              | 5.0    |
| dtype: float64     |        |

```
[565]: data_num.mode().sum()
```

|                    |        |
|--------------------|--------|
| Inventory Level    | 159.0  |
| Units Sold         | 40.0   |
| Units Ordered      | 68.0   |
| Demand Forecast    | 28.0   |
| Price              | 92.0   |
| Discount           | 15.0   |
| Holiday/Promotion  | 1.0    |
| Competitor Pricing | 42.0   |
| Year               | 2022.0 |
| Month              | 4.0    |
| dtype: float64     |        |

```
[567]: data_num.std()
```

|                    |            |
|--------------------|------------|
| Inventory Level    | 129.727003 |
| Units Sold         | 108.938973 |
| Units Ordered      | 52.333915  |
| Demand Forecast    | 109.290107 |
| Price              | 26.038579  |
| Discount           | 7.077242   |
| Holiday/Promotion  | 0.500005   |
| Competitor Pricing | 26.201560  |
| Year               | 0.440266   |
| Month              | 3.468399   |
| dtype: float64     |            |

## 16 Formula for Variance:

- For a dataset with  $n$  observations  $1, 2, \dots, n$ , the variance is calculated as:

$$\text{Variance} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

where  $\bar{x}$  is the mean of the dataset.

```
[570]: data_num.var()
```

|                 |              |
|-----------------|--------------|
| Inventory Level | 16829.095335 |
| Units Sold      | 11867.699941 |

|                    |              |
|--------------------|--------------|
| Units Ordered      | 2738.838637  |
| Demand Forecast    | 11944.327589 |
| Price              | 678.007614   |
| Discount           | 50.087361    |
| Holiday/Promotion  | 0.250005     |
| Competitor Pricing | 686.521770   |
| Year               | 0.193834     |
| Month              | 12.029794    |

dtype: float64

## 17 Steps to Find the Range

- **Find the Maximum Value:** Use the `.max()` method.
- **Find the Minimum Value:** Use the `.min()` method.
- **Calculate the Range:** Subtract the minimum value from the maximum value.

### Method#1

```
[574]: # Calculate range for the 'Price' column
price_range = data['Price'].max() - data['Price'].min()

# Calculate range for the 'Inventory Level' column
inv_range = data['Inventory Level'].max() - data['Inventory Level'].min()

print(f"Range of Price: {price_range}")
print(f"Range of Inventory Level: {inv_range}")
```

Range of Price: 90

Range of Inventory Level: 450

### Method#2

```
[577]: # Apply a function to calculate the range for each column
range_per_column = data.select_dtypes(include=['number']).apply(lambda x: x.
    ↪max() - x.min())

print(range_per_column)
```

|                    |     |
|--------------------|-----|
| Inventory Level    | 450 |
| Units Sold         | 496 |
| Units Ordered      | 180 |
| Demand Forecast    | 521 |
| Price              | 90  |
| Discount           | 20  |
| Holiday/Promotion  | 1   |
| Competitor Pricing | 100 |
| Year               | 1   |

```
Month
dtype: int64
```

```
[579]: data_num.cov()          ## Covariance of numeric columns of a dataset
```

```
[579]:
```

|                    | Inventory Level | Units Sold   | Units Ordered | \ |
|--------------------|-----------------|--------------|---------------|---|
| Inventory Level    | 16829.095335    | 8330.343251  | 8.565579      |   |
| Units Sold         | 8330.343251     | 11867.699941 | -21.486230    |   |
| Units Ordered      | 8.565579        | -21.486230   | 2738.838637   |   |
| Demand Forecast    | 8336.634711     | 11868.862558 | -24.306124    |   |
| Price              | 37.012100       | 0.766777     | 9.487898      |   |
| Discount           | 4.891312        | 1.589120     | -0.682090     |   |
| Holiday/Promotion  | 0.392525        | 0.293232     | 0.107420      |   |
| Competitor Pricing | 36.366017       | 0.513854     | 9.528475      |   |
| Year               | -0.113468       | -0.270901    | 0.132831      |   |
| Month              | 1.745196        | -0.064008    | -1.055695     |   |

|                    | Demand Forecast | Price      | Discount  | Holiday/Promotion | \ |
|--------------------|-----------------|------------|-----------|-------------------|---|
| Inventory Level    | 8336.634711     | 37.012100  | 4.891312  | 0.392525          |   |
| Units Sold         | 11868.862558    | 0.766777   | 1.589120  | 0.293232          |   |
| Units Ordered      | -24.306124      | 9.487898   | -0.682090 | 0.107420          |   |
| Demand Forecast    | 11944.327589    | -0.641965  | 1.454638  | 0.279938          |   |
| Price              | -0.641965       | 678.007614 | 0.263326  | 0.004153          |   |
| Discount           | 1.454638        | 0.263326   | 50.087361 | -0.004241         |   |
| Holiday/Promotion  | 0.279938        | 0.004153   | -0.004241 | 0.250005          |   |
| Competitor Pricing | -0.884766       | 677.979257 | 0.283088  | -0.003632         |   |
| Year               | -0.267948       | 0.039208   | 0.004462  | 0.000805          |   |
| Month              | -0.152368       | -0.505880  | -0.056446 | 0.004840          |   |

|                    | Competitor Pricing | Year      | Month     |
|--------------------|--------------------|-----------|-----------|
| Inventory Level    | 36.366017          | -0.113468 | 1.745196  |
| Units Sold         | 0.513854           | -0.270901 | -0.064008 |
| Units Ordered      | 9.528475           | 0.132831  | -1.055695 |
| Demand Forecast    | -0.884766          | -0.267948 | -0.152368 |
| Price              | 677.979257         | 0.039208  | -0.505880 |
| Discount           | 0.283088           | 0.004462  | -0.056446 |
| Holiday/Promotion  | -0.003632          | 0.000805  | 0.004840  |
| Competitor Pricing | 686.521770         | 0.049073  | -0.510193 |
| Year               | 0.049073           | 0.193834  | -0.742247 |
| Month              | -0.510193          | -0.742247 | 12.029794 |

## 18 5. Data Visualization

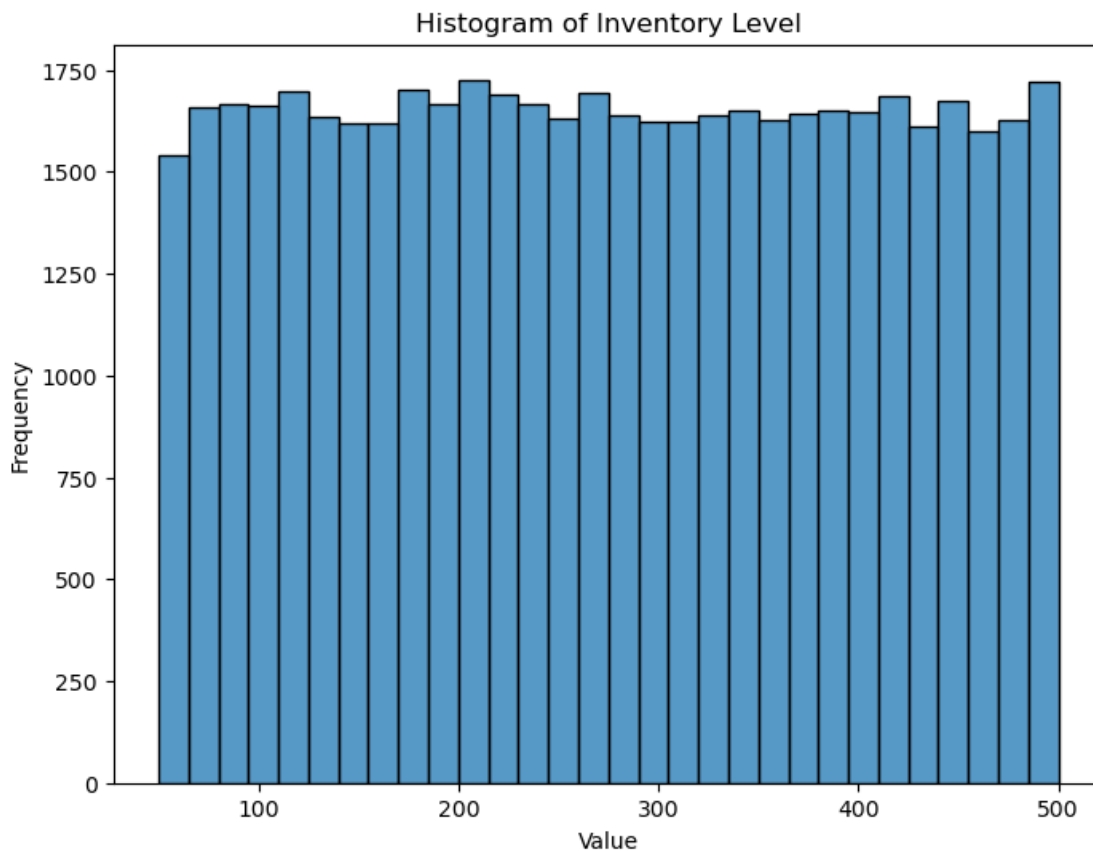
- **Univariate Analysis:**
  - **Numerical Variables:** Histograms, boxplots, density plots.
  - **Categorical Variables:** Bar charts, count plots.

## 18.1 Tools: matplotlib, seaborn, pandas.plot().

- **Multivariate Analysis:**

- **Correlation Heatmaps:** Identify relationships using `seaborn.heatmap()`.
- **Scatter Plots:** Examine pairwise relationships.
- **Pairplot:** Use `seaborn.pairplot()` for quick visualization of numeric variables.

```
[582]: # Plot histogram
plt.figure(figsize=(8, 6))
sns.histplot(data['Inventory Level'], kde=False, bins=30) # kde=False removes
↳ the density curve
plt.title("Histogram of Inventory Level")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```

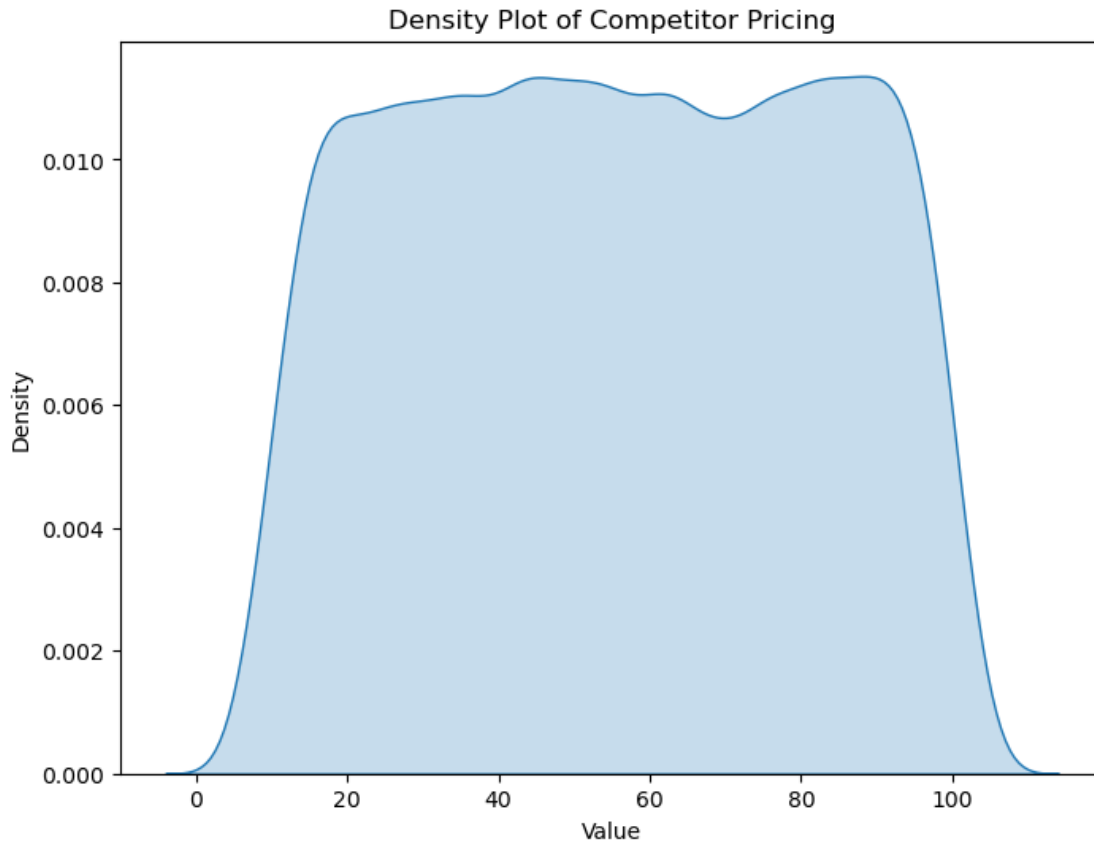


```
[584]: import warnings
warnings.filterwarnings("ignore")
plt.figure(figsize=(8, 6))
```

```

sns.kdeplot(data['Competitor Pricing'], shade=True)
plt.title("Density Plot of Competitor Pricing")
plt.xlabel("Value")
plt.ylabel("Density")
plt.show()

```



```

[586]: plt.figure(figsize=(15,7))
ax = sns.countplot(x=data['Name of Month'],orient='v')
for p in ax.patches:
    ax.annotate(f'{p.get_height()}',
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='center',
                fontsize=12, color='black',
                xytext=(0, 10), textcoords='offset points')
plt.xticks(rotation=30)

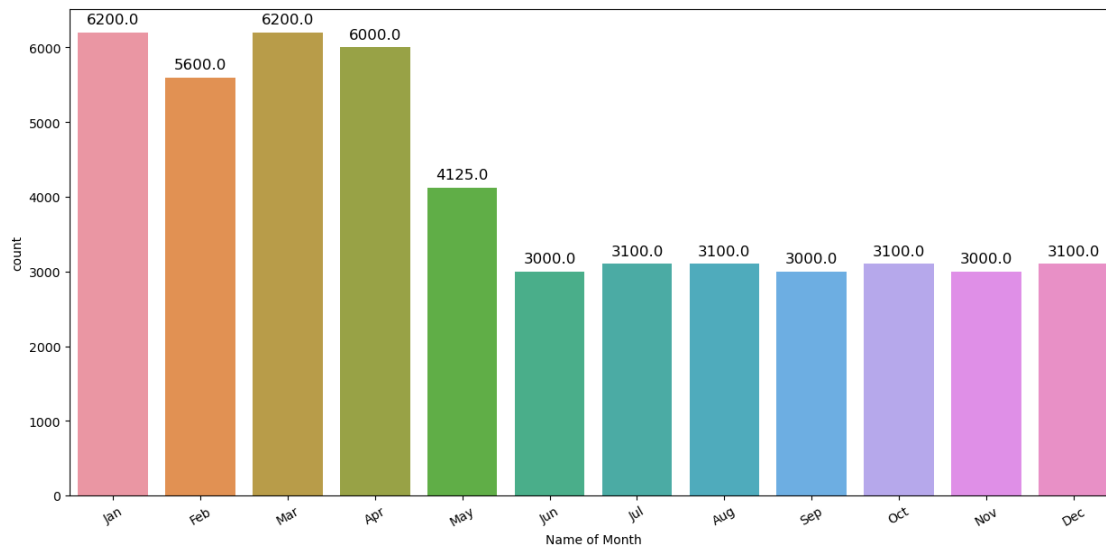
```

```

[586]: (array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11]),
       [Text(0, 0, 'Jan'),
        Text(1, 0, 'Feb'),
        Text(2, 0, 'Mar'),

```

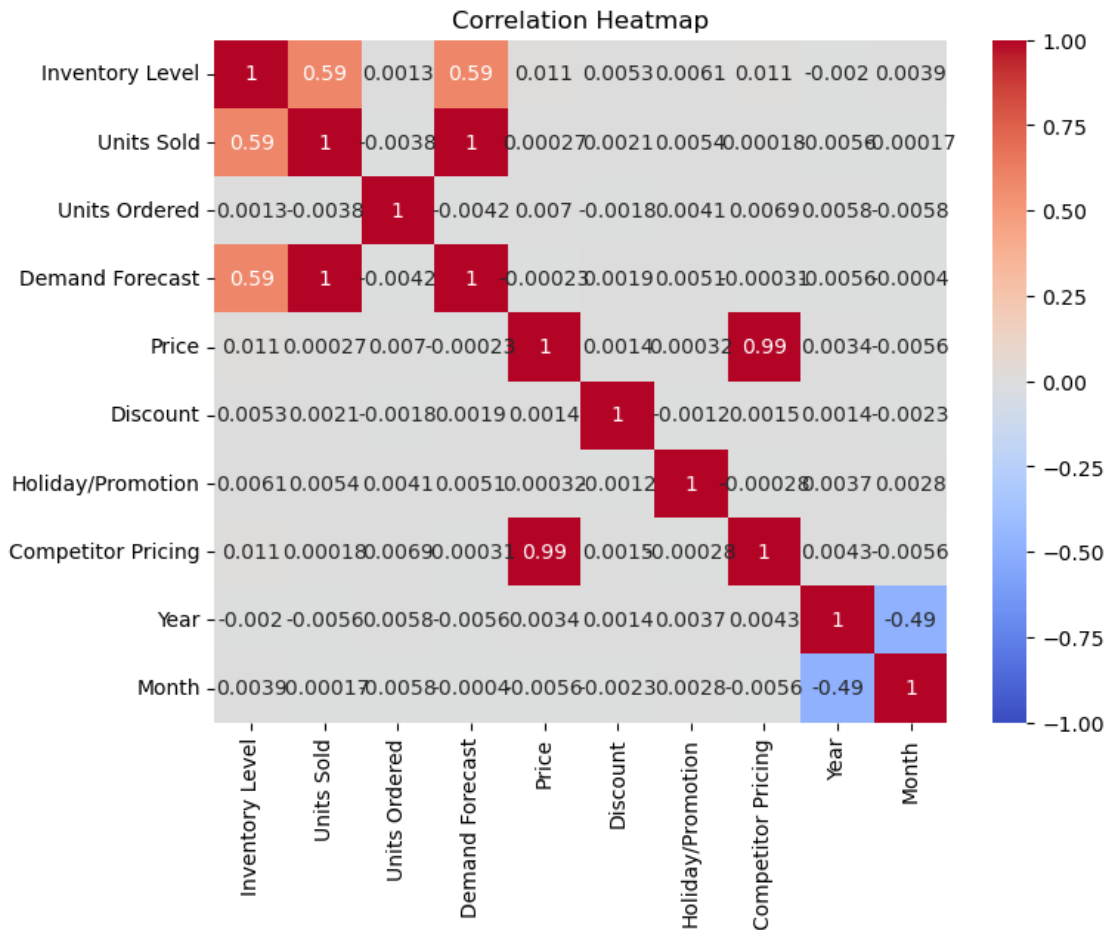
```
Text(3, 0, 'Apr'),
Text(4, 0, 'May'),
Text(5, 0, 'Jun'),
Text(6, 0, 'Jul'),
Text(7, 0, 'Aug'),
Text(8, 0, 'Sep'),
Text(9, 0, 'Oct'),
Text(10, 0, 'Nov'),
Text(11, 0, 'Dec'))]
```



[588]: *#A correlation heatmap is a great way to visually identify relationships  
 ↳ between numerical variables. It shows the correlation coefficients (values  
 ↳ between -1 and 1) between each pair of variables in the dataset.*

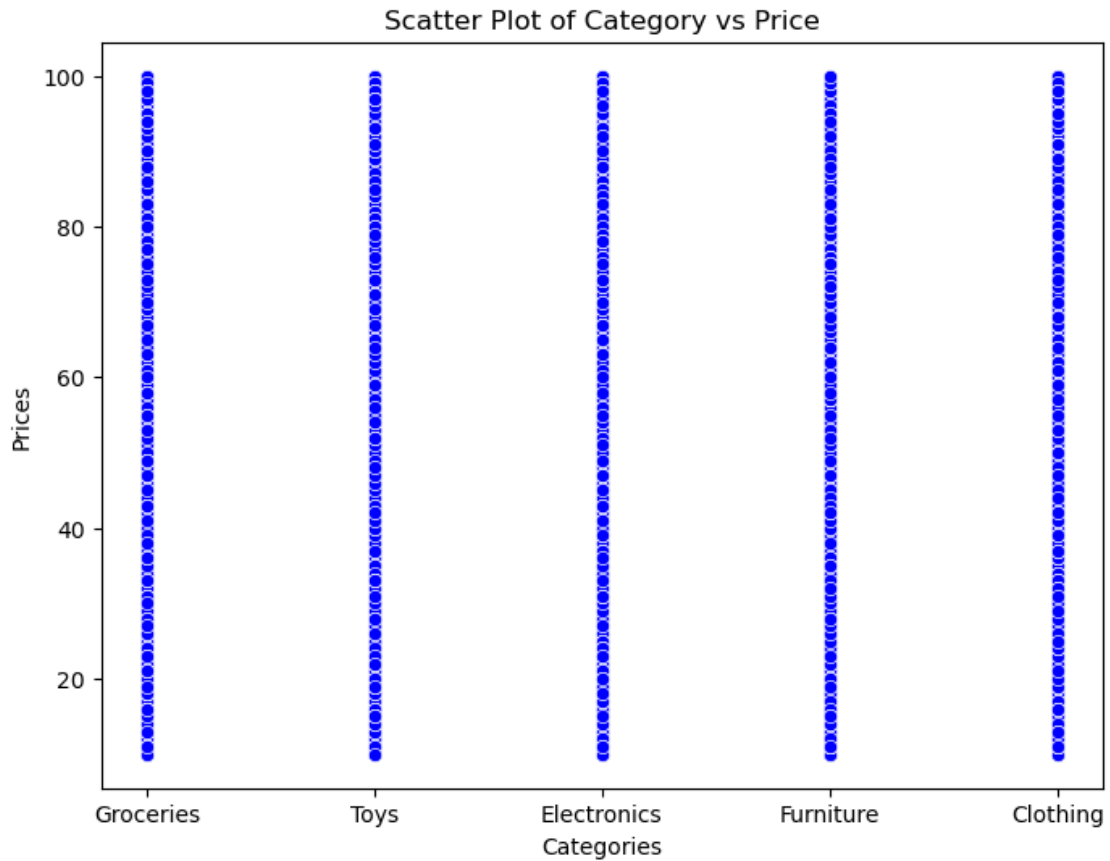
```
# Compute the correlation matrix
corr = data_num.corr()

# Plot the heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap="coolwarm", vmin=-1, vmax=1)
plt.title("Correlation Heatmap")
plt.show()
```



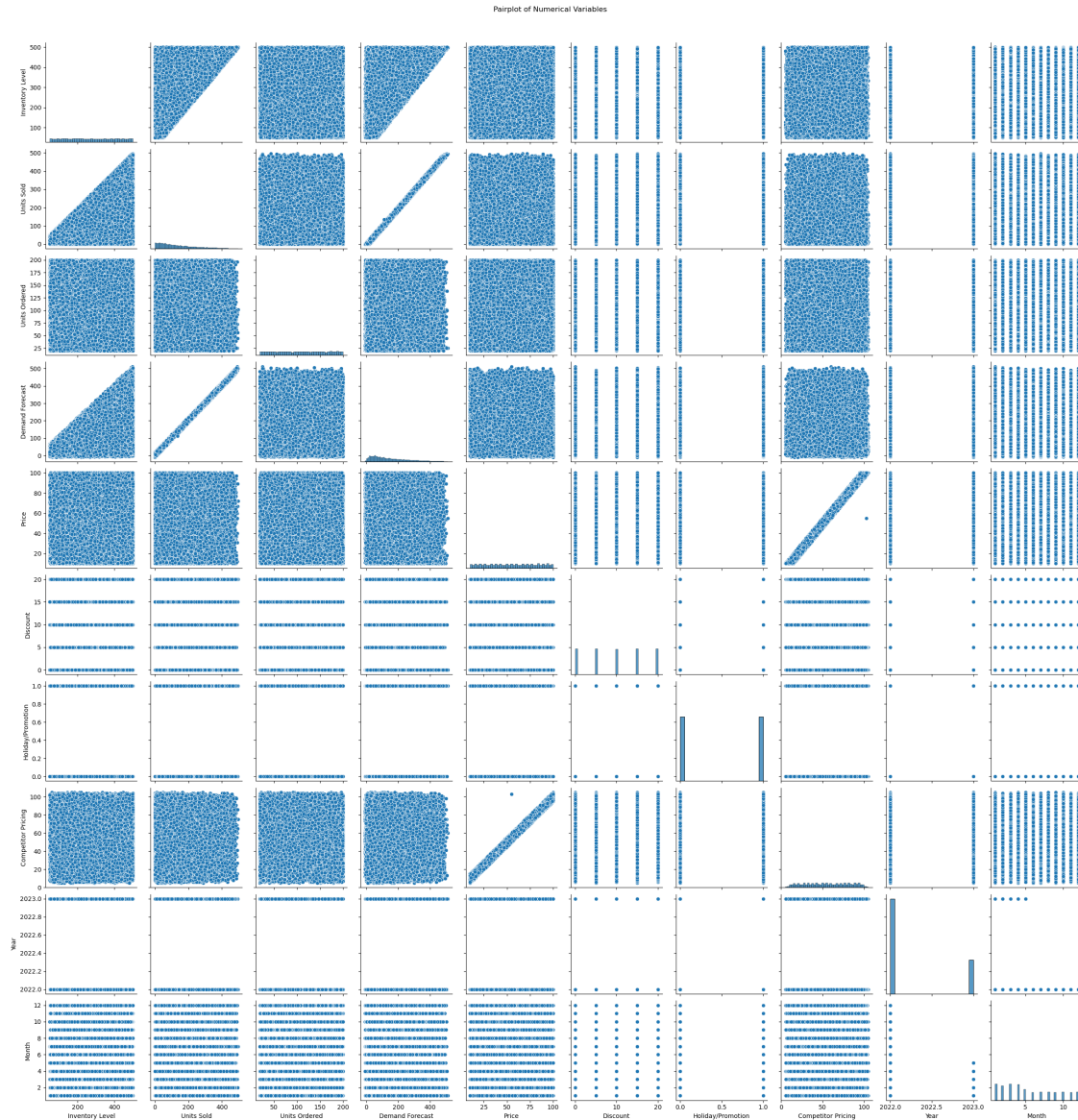
```
[590]: #A scatter plot is used to examine the relationship between two continuous
        ↪ variables. It helps identify patterns, trends, or outliers.

        # Scatter plot for two numerical variables
        plt.figure(figsize=(8, 6))
        sns.scatterplot(x=data['Category'], y=data['Price'], data=data, color='blue')
        plt.title("Scatter Plot of Category vs Price")
        plt.xlabel("Categories")
        plt.ylabel("Prices")
        plt.show()
```



[592]: *#A pairplot provides a quick way to visualize pairwise relationships between*  
*↪ multiple numerical variables. It automatically creates scatter plots for all*  
*↪ pairs of variables and histograms on the diagonal.*

```
# Pairplot for visualizing relationships between all numerical variables  
sns.pairplot(data)  
plt.suptitle("Pairplot of Numerical Variables", y=1.02)  
plt.show()
```



[ ]:

## 19 6. Identify Outliers

- **Boxplots:** Identify outliers visually.
- **IQR Method:** Detect outliers using the Interquartile Range:

$$- \text{Outliers: } < 1 - 1.5 \times \text{IQR}$$

$$\text{or } > 3 + 1.5 \times \text{IQR}$$

- **Z-Scores:** Use `scipy.stats.zscore()` to find outliers.

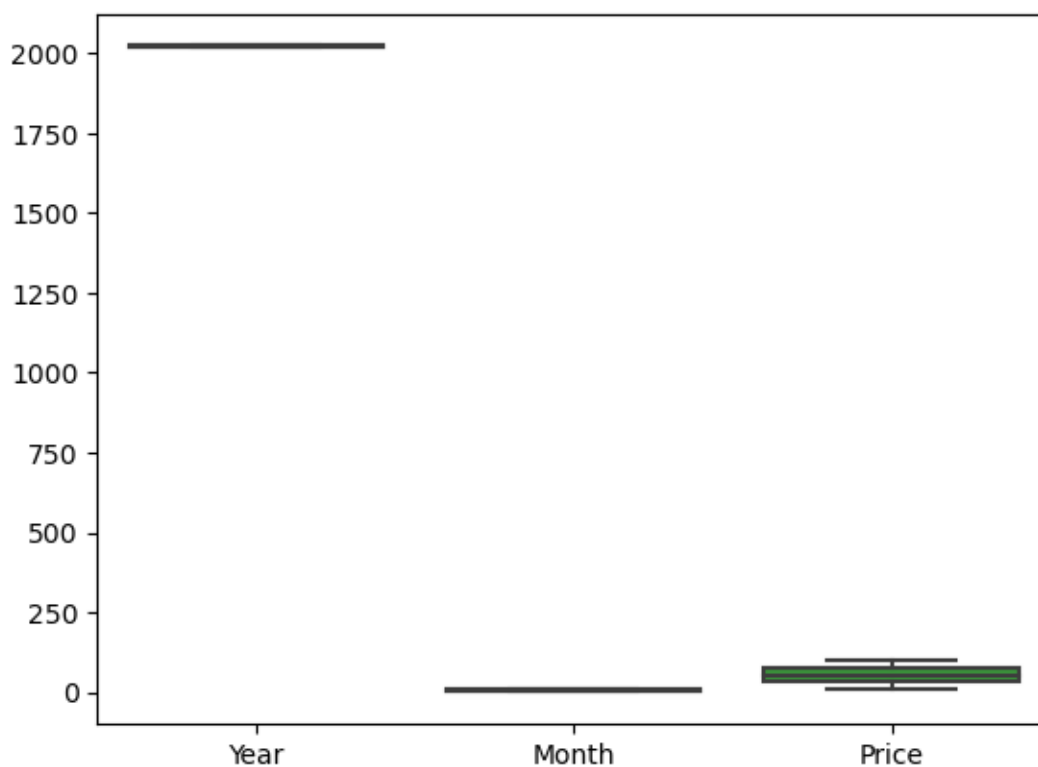
```
[217]: def wisker(col):  
        q1, q3 = np.percentile(col, [25, 75])  
        iqr = q3 - q1  
        lw = q1 - 1.5*iqr  
        up = q3 + 1.5*iqr  
        return lw, up
```

```
[219]: wisker(data[['Year', 'Month', 'Price']])
```

```
[219]: (-3013.0, 5043.0)
```

```
[221]: for i in [['Year', 'Month', 'Price']]:  
        lw, up = wisker(data[i])  
        data[i] = np.where(data[i] < lw, lw, data[i])  
        data[i] = np.where(data[i] > up, up, data[i])
```

```
[223]: for i in [['Year', 'Month', 'Price']]:  
        sns.boxplot(data[i])  
        plt.show()
```

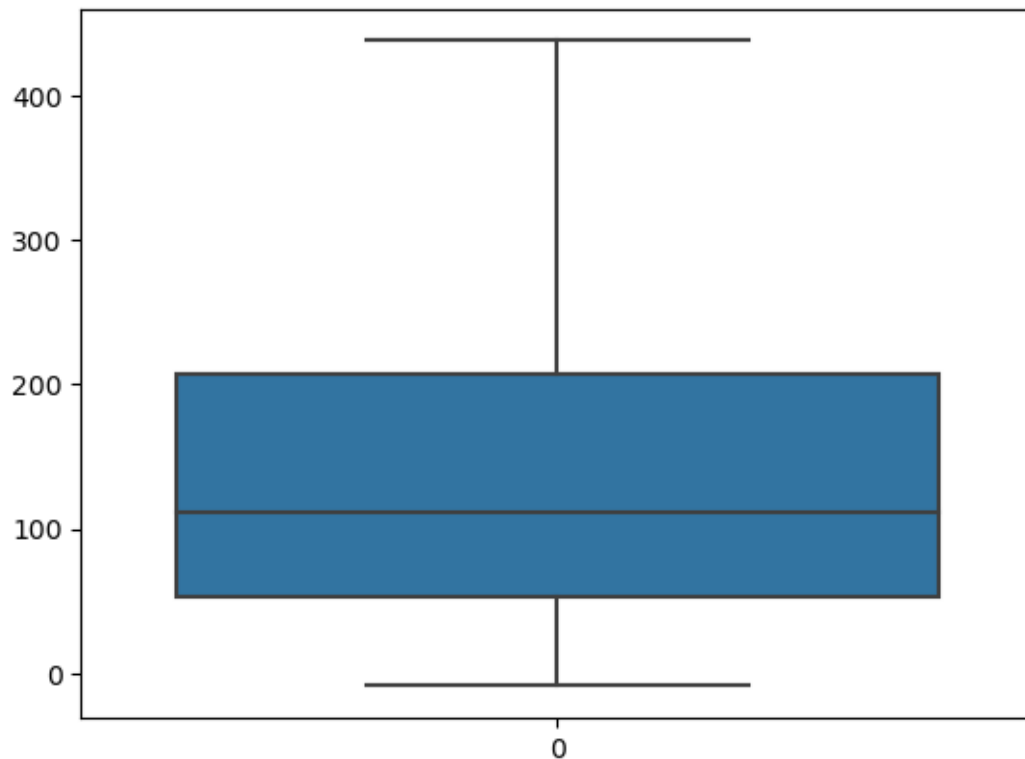


```
[225]: wisker(data['Demand Forecast'])
```

```
[225]: (-178.0, 438.0)
```

```
[227]: for i in ['Demand Forecast']:
        lw, up = wisker(data[i])
        data[i] = np.where(data[i]<lw, lw, data[i])
        data[i] = np.where(data[i]>up, up, data[i])
```

```
[229]: for i in ['Demand Forecast']:
        sns.boxplot(data[i])
        plt.show()
```



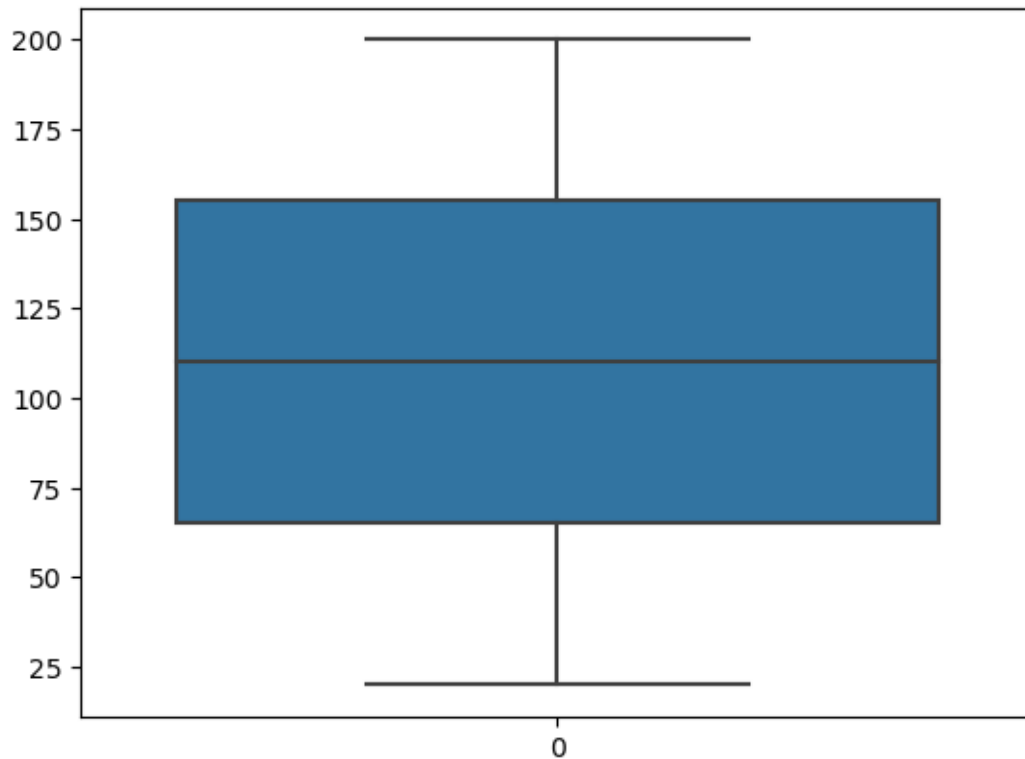
```
[231]: wisker(data['Units Ordered'])
```

```
[231]: (-70.0, 290.0)
```

```
[233]: for i in ['Units Ordered']:
        lw, up = wisker(data[i])
        data[i] = np.where(data[i]<lw, lw, data[i])
        data[i] = np.where(data[i]>up, up, data[i])
```

```
[235]: for i in ['Units Ordered']:
        sns.boxplot(data[i])
```

```
plt.show()
```



```
[237]: from scipy.stats import zscore
# Calculate Z-scores
data['Z-Score'] = zscore(data['Inventory Level'])

# Identify outliers: Values with a Z-score greater than 3 or less than -3 are
↳ outliers
outliers = data[data['Z-Score'].abs() > 3]

# Show the Z-scores and the outliers
print(data)
print("\nOutliers based on Z-score:")
print(outliers)
```

|     | Date       | Store ID | Product ID | Category    | Region | Inventory Level \ |
|-----|------------|----------|------------|-------------|--------|-------------------|
| 0   | 2022-01-01 | S001     | P0001      | Groceries   | North  | 231               |
| 1   | 2022-01-01 | S001     | P0002      | Toys        | South  | 204               |
| 2   | 2022-01-01 | S001     | P0003      | Toys        | West   | 102               |
| 3   | 2022-01-01 | S001     | P0004      | Toys        | North  | 469               |
| 4   | 2022-01-01 | S001     | P0005      | Electronics | East   | 166               |
| ... | ...        | ...      | ...        | ...         | ...    | ...               |

|       |            |      |       |           |       |     |
|-------|------------|------|-------|-----------|-------|-----|
| 49520 | 2023-05-11 | S002 | P0001 | Clothing  | West  | 385 |
| 49521 | 2023-05-11 | S002 | P0002 | Toys      | South | 443 |
| 49522 | 2023-05-11 | S002 | P0003 | Groceries | North | 493 |
| 49523 | 2023-05-11 | S002 | P0004 | Furniture | West  | 498 |
| 49524 | 2023-05-11 | S002 | P000  | Groceries | West  | 274 |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount \ |
|-------|------------|---------------|-----------------|-------|------------|
| 0     | 127        | 55.0          | 135.0           | 34.0  | 20         |
| 1     | 150        | 66.0          | 144.0           | 63.0  | 20         |
| 2     | 65         | 51.0          | 74.0            | 28.0  | 10         |
| 3     | 61         | 164.0         | 62.0            | 33.0  | 10         |
| 4     | 14         | 135.0         | 9.0             | 74.0  | 0          |
| ...   | ...        | ...           | ...             | ...   | ...        |
| 49520 | 143        | 36.0          | 143.0           | 42.0  | 15         |
| 49521 | 77         | 53.0          | 79.0            | 33.0  | 20         |
| 49522 | 100        | 23.0          | 94.0            | 21.0  | 10         |
| 49523 | 461        | 197.0         | 438.0           | 100.0 | 20         |
| 49524 | 136        | 110.0         | 112.0           | 55.0  | 10         |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality \ |
|-------|-------------------|-------------------|--------------------|---------------|
| 0     | Rainy             | 0                 | 30                 | Autumn        |
| 1     | Sunny             | 0                 | 66                 | Autumn        |
| 2     | Sunny             | 1                 | 31                 | Summer        |
| 3     | Cloudy            | 1                 | 35                 | Autumn        |
| 4     | Sunny             | 0                 | 69                 | Summer        |
| ...   | ...               | ...               | ...                | ...           |
| 49520 | Sunny             | 0                 | 47                 | Spring        |
| 49521 | Cloudy            | 0                 | 32                 | Winter        |
| 49522 | Snowy             | 1                 | 20                 | Spring        |
| 49523 | Snowy             | 0                 | 103                | Winter        |
| 49524 | Snowy             | 1                 | 103                | Spring        |

|       | Year   | Month Name of Month | Z-Score   |
|-------|--------|---------------------|-----------|
| 0     | 2022.0 | 1.0 Jan             | -0.336377 |
| 1     | 2022.0 | 1.0 Jan             | -0.544508 |
| 2     | 2022.0 | 1.0 Jan             | -1.330783 |
| 3     | 2022.0 | 1.0 Jan             | 1.498263  |
| 4     | 2022.0 | 1.0 Jan             | -0.837434 |
| ...   | ...    | ...                 | ...       |
| 49520 | 2023.0 | 5.0 May             | 0.850743  |
| 49521 | 2023.0 | 5.0 May             | 1.297841  |
| 49522 | 2023.0 | 5.0 May             | 1.683269  |
| 49523 | 2023.0 | 5.0 May             | 1.721812  |
| 49524 | 2023.0 | 5.0 May             | -0.004908 |

[49525 rows x 19 columns]

Outliers based on Z-score:

Empty DataFrame

Columns: [Date, Store ID, Product ID, Category, Region, Inventory Level, Units Sold, Units Ordered, Demand Forecast, Price, Discount, Weather Condition, Holiday/Promotion, Competitor Pricing, Seasonality, Year, Month, Name of Month, Z-Score]

Index: []

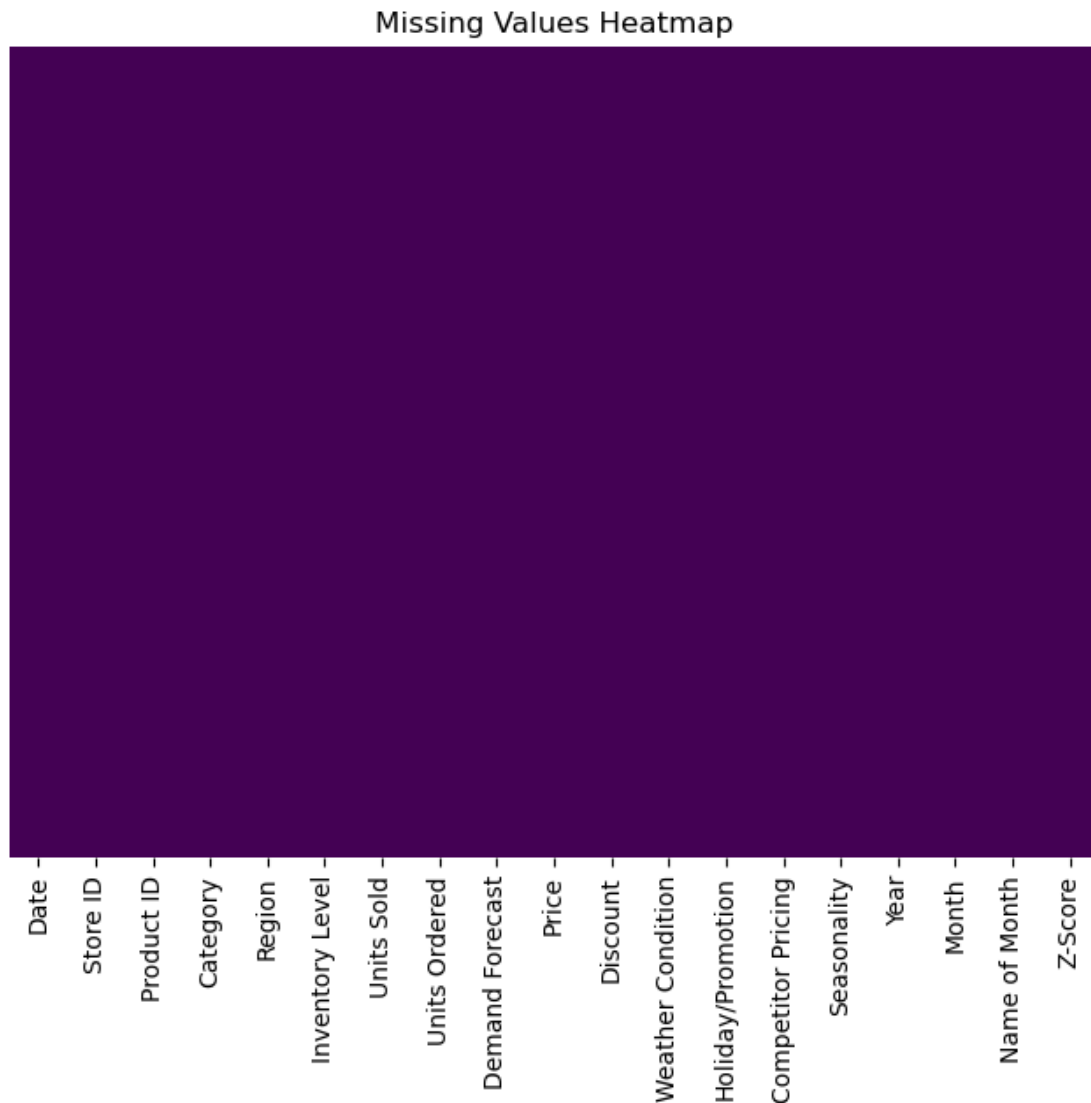
## 20 7. Missing Data Analysis

- **Missing Value Counts:** `data.isnull().sum()`
- **Heatmap:** Visualize missing values using `seaborn.heatmap(data.isnull(), cbar=False)`.

```
[239]: data.isnull().sum()
```

```
[239]: Date                0
Store ID                0
Product ID             0
Category               0
Region                0
Inventory Level        0
Units Sold             0
Units Ordered          0
Demand Forecast        0
Price                 0
Discount              0
Weather Condition      0
Holiday/Promotion      0
Competitor Pricing     0
Seasonality           0
Year                  0
Month                 0
Name of Month          0
Z-Score               0
dtype: int64
```

```
[241]: plt.figure(figsize=(8, 6))
sns.heatmap(data.isnull(), cbar=False, cmap="viridis", yticklabels=False,
            xticklabels=data.columns)
plt.title("Missing Values Heatmap")
plt.show()
```



## 21 8. Relationship Exploration

- **Correlation Analysis:**
  - Use `data.corr()` for Pearson correlation.
- **Cross-tabulation:** For categorical variables using `pandas.crosstab()`.

```
[243]: data_num.corr() ## Correlation of a dataset
```

```
[243]:
```

|                 | Inventory Level | Units Sold | Units Ordered \ |
|-----------------|-----------------|------------|-----------------|
| Inventory Level | 1.000000        | 0.589453   | 0.001262        |
| Units Sold      | 0.589453        | 1.000000   | -0.003769       |

|                    |           |           |           |
|--------------------|-----------|-----------|-----------|
| Units Ordered      | 0.001262  | -0.003769 | 1.000000  |
| Demand Forecast    | 0.588003  | 0.996885  | -0.004250 |
| Price              | 0.010957  | 0.000270  | 0.006963  |
| Discount           | 0.005328  | 0.002061  | -0.001842 |
| Holiday/Promotion  | 0.006051  | 0.005383  | 0.004105  |
| Competitor Pricing | 0.010699  | 0.000180  | 0.006949  |
| Year               | -0.001987 | -0.005648 | 0.005765  |
| Month              | 0.003879  | -0.000169 | -0.005816 |

|                    | Demand Forecast | Price     | Discount  | Holiday/Promotion \ |
|--------------------|-----------------|-----------|-----------|---------------------|
| Inventory Level    | 0.588003        | 0.010957  | 0.005328  | 0.006051            |
| Units Sold         | 0.996885        | 0.000270  | 0.002061  | 0.005383            |
| Units Ordered      | -0.004250       | 0.006963  | -0.001842 | 0.004105            |
| Demand Forecast    | 1.000000        | -0.000226 | 0.001881  | 0.005123            |
| Price              | -0.000226       | 1.000000  | 0.001429  | 0.000319            |
| Discount           | 0.001881        | 0.001429  | 1.000000  | -0.001199           |
| Holiday/Promotion  | 0.005123        | 0.000319  | -0.001199 | 1.000000            |
| Competitor Pricing | -0.000309       | 0.993738  | 0.001527  | -0.000277           |
| Year               | -0.005569       | 0.003420  | 0.001432  | 0.003655            |
| Month              | -0.000402       | -0.005601 | -0.002300 | 0.002791            |

|                    | Competitor Pricing | Year      | Month     |
|--------------------|--------------------|-----------|-----------|
| Inventory Level    | 0.010699           | -0.001987 | 0.003879  |
| Units Sold         | 0.000180           | -0.005648 | -0.000169 |
| Units Ordered      | 0.006949           | 0.005765  | -0.005816 |
| Demand Forecast    | -0.000309          | -0.005569 | -0.000402 |
| Price              | 0.993738           | 0.003420  | -0.005601 |
| Discount           | 0.001527           | 0.001432  | -0.002300 |
| Holiday/Promotion  | -0.000277          | 0.003655  | 0.002791  |
| Competitor Pricing | 1.000000           | 0.004254  | -0.005614 |
| Year               | 0.004254           | 1.000000  | -0.486076 |
| Month              | -0.005614          | -0.486076 | 1.000000  |

```
[245]: from scipy.stats import chi2_contingency

contingency_table = pd.crosstab(data['Year'], data['Region'])

# Perform the chi-square test
chi2, p, dof, expected = chi2_contingency(contingency_table)

# Output the results
print("Contingency Table:")
print(contingency_table)
print("\nChi-Square Statistic:", chi2)
print("P-Value:", p)
print("Degrees of Freedom:", dof)
print("\nExpected Frequencies:")
```

```

print(expected)

# Decision based on p-value
if p <= 0.05:
    print("\nReject the null hypothesis: Variables are associated.")
else:
    print("\nFail to reject the null hypothesis: No association.")

```

Contingency Table:

| Region | East | North | South | West |
|--------|------|-------|-------|------|
| Year   |      |       |       |      |
| 2022.0 | 9164 | 9122  | 9104  | 9110 |
| 2023.0 | 3225 | 3267  | 3231  | 3302 |

Chi-Square Statistic: 1.1542862860096779

P-Value: 0.7639879036763937

Degrees of Freedom: 3

Expected Frequencies:

```

[[9130.71176174 9130.71176174 9090.91367996 9147.66279657]
 [3258.28823826 3258.28823826 3244.08632004 3264.33720343]]

```

Fail to reject the null hypothesis: No association.

```

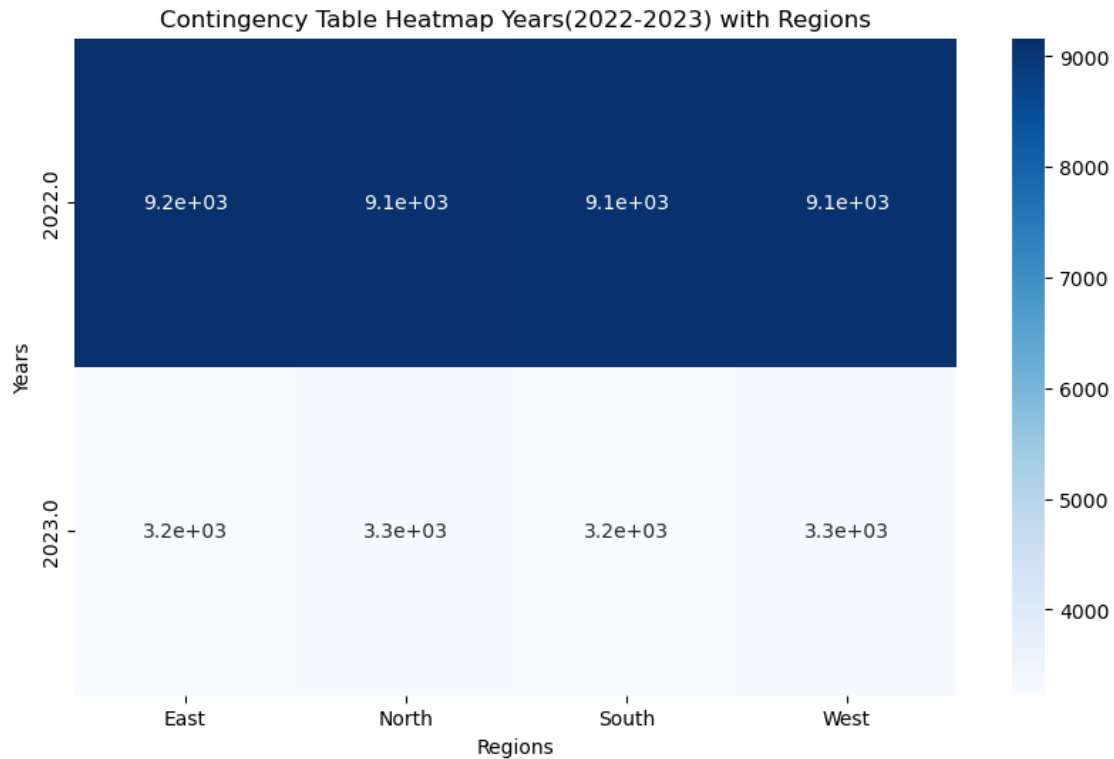
[247]: subset_data = contingency_table.loc[2022:2023, ['East', 'North', 'South', 'West']]

```

```

# Plot heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(subset_data, annot=True, cmap="Blues")
plt.title("Contingency Table Heatmap Years(2022-2023) with Regions")
plt.xlabel("Regions")
plt.ylabel("Years")
plt.show()

```



```
[249]: contingency_table = pd.crosstab(data['Weather Condition'], data['Seasonality'])

# Perform the chi-square test
chi2, p, dof, expected = chi2_contingency(contingency_table)

# Output the results
print("Contingency Table:")
print(contingency_table)
print("\nChi-Square Statistic:", chi2)
print("P-Value:", p)
print("Degrees of Freedom:", dof)
print("\nExpected Frequencies:")
print(expected)

# Decision based on p-value
if p <= 0.05:
    print("\nReject the null hypothesis: Variables are associated.")
else:
    print("\nFail to reject the null hypothesis: No association.")
```

```
Contingency Table:
Seasonality      Autumn  Spring  Summer  Winter
Weather Condition
```

|        |      |      |      |      |
|--------|------|------|------|------|
| Cloudy | 3006 | 3217 | 3057 | 3081 |
| Rainy  | 3124 | 3057 | 3100 | 3089 |
| Snowy  | 3132 | 3018 | 3175 | 3083 |
| Sunny  | 3113 | 3151 | 3045 | 3077 |

Chi-Square Statistic: 14.492963629911928

P-Value: 0.10583841872634041

Degrees of Freedom: 9

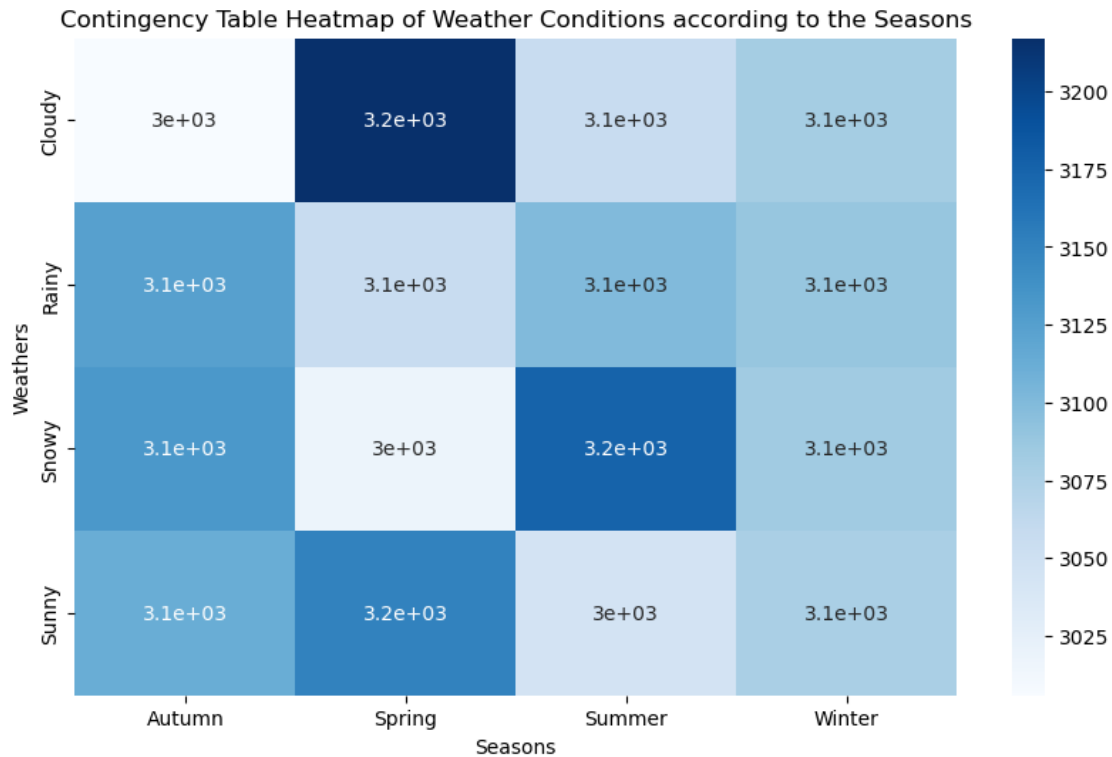
Expected Frequencies:

```
[[3088.69005553 3105.66225139 3089.18923776 3077.45845533]
 [3090.93891974 3107.92347299 3091.43846542 3079.69914185]
 [3100.43412418 3117.4708531 3100.93520444 3089.15981827]
 [3094.93690056 3111.94342251 3095.43709238 3083.68258455]]
```

Fail to reject the null hypothesis: No association.

```
[251]: subset_data = contingency_table.loc[ : , : ]

# Plot heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(subset_data, annot=True, cmap="Blues")
plt.title("Contingency Table Heatmap of Weather Conditions according to the_
↪Seasons")
plt.ylabel("Weathers")
plt.xlabel("Seasons")
plt.show()
```



```
[253]: contingency_table = pd.crosstab(data['Category'], data['Inventory Level'])
```

```
# Perform the chi-square test
chi2, p, dof, expected = chi2_contingency(contingency_table)

# Output the results
print("Contingency Table:")
print(contingency_table)
print("\nChi-Square Statistic:", chi2)
print("P-Value:", p)
print("Degrees of Freedom:", dof)
print("\nExpected Frequencies:")
print(expected)

# Decision based on p-value
if p <= 0.05:
    print("\nReject the null hypothesis: Variables are associated.")
else:
    print("\nFail to reject the null hypothesis: No association.")
```

Contingency Table:

|                 |    |    |    |    |    |    |    |    |    |    |     |     |   |
|-----------------|----|----|----|----|----|----|----|----|----|----|-----|-----|---|
| Inventory Level | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | ... | 491 | \ |
| Category        |    |    |    |    |    |    |    |    |    |    |     | ... |   |

|             |    |    |    |    |    |    |    |    |    |    |     |    |
|-------------|----|----|----|----|----|----|----|----|----|----|-----|----|
| Clothing    | 18 | 16 | 18 | 20 | 21 | 20 | 23 | 22 | 28 | 21 | ... | 12 |
| Electronics | 16 | 15 | 14 | 13 | 26 | 30 | 26 | 21 | 21 | 21 | ... | 18 |
| Furniture   | 16 | 26 | 22 | 15 | 24 | 21 | 14 | 25 | 29 | 13 | ... | 19 |
| Groceries   | 20 | 18 | 21 | 25 | 19 | 22 | 19 | 22 | 15 | 20 | ... | 21 |
| Toys        | 17 | 20 | 14 | 19 | 19 | 19 | 21 | 22 | 24 | 33 | ... | 25 |

| Inventory Level | 492 | 493 | 494 | 495 | 496 | 497 | 498 | 499 | 500 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Category        |     |     |     |     |     |     |     |     |     |
| Clothing        | 23  | 17  | 17  | 24  | 20  | 19  | 23  | 18  | 18  |
| Electronics     | 26  | 24  | 19  | 22  | 24  | 19  | 24  | 23  | 24  |
| Furniture       | 15  | 17  | 28  | 25  | 24  | 13  | 14  | 28  | 22  |
| Groceries       | 26  | 28  | 31  | 23  | 23  | 18  | 21  | 19  | 25  |
| Toys            | 27  | 18  | 19  | 26  | 24  | 19  | 26  | 19  | 19  |

[5 rows x 451 columns]

Chi-Square Statistic: 1800.330391992226

P-Value: 0.49337090785102405

Degrees of Freedom: 1800

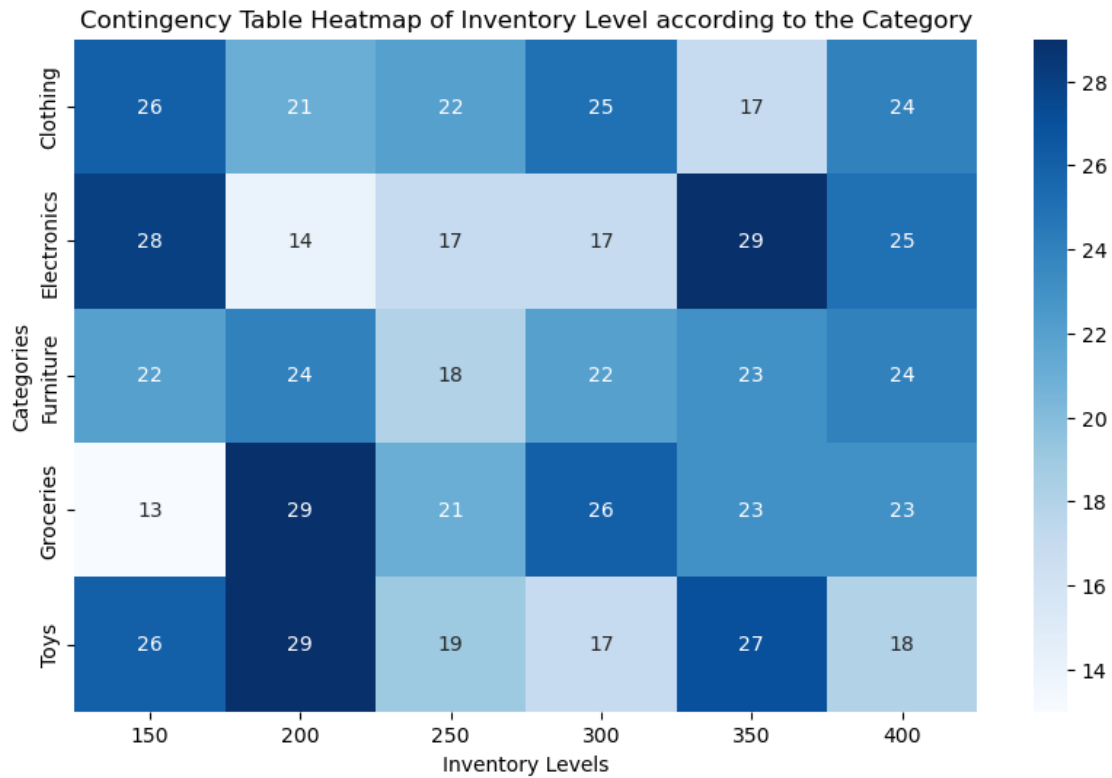
Expected Frequencies:

```
[17.42283695 19.0249369 17.82336194 ... 21.62834932 21.42808682
 21.62834932]
[17.08203937 18.65280162 17.47472993 ... 21.20529026 21.00894498
 21.20529026]
[17.38243311 18.98081777 17.78202928 ... 21.57819283 21.37839475
 21.57819283]
[17.64066633 19.26279657 18.04619889 ... 21.8987582 21.69599192
 21.8987582 ]
[17.47202423 19.07864715 17.87367996 ... 21.68940939 21.48858152
 21.68940939]]
```

Fail to reject the null hypothesis: No association.

```
[255]: subset_data = contingency_table.loc[ : , [150,200,250,300,350,400]]

# Plot heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(subset_data, annot=True, cmap="Blues")
plt.title("Contingency Table Heatmap of Inventory Level according to the_
↳Category")
plt.ylabel("Categories")
plt.xlabel("Inventory Levels")
plt.show()
```



## 22 9. Categorical Data Analysis

- **Frequency Counts:** `data['column'].value_counts()`
- **Proportions:** Use percentages for categorical distributions.
- **Chi-Square Tests:** Assess relationships between categorical variables.

```
[257]: data['Inventory Level'].value_counts()
```

```
[257]: Inventory Level
159    143
268    142
387    142
429    140
231    138
...
497      88
160      87
50       87
489      87
165      77
Name: count, Length: 451, dtype: int64
```

```
[259]: data['Category'].value_counts()
```

```
[259]: Category
Groceries      10042
Toys           9946
Clothing       9918
Furniture      9895
Electronics    9724
Name: count, dtype: int64
```

```
[261]: data['Name of Month'].value_counts()
```

```
[261]: Name of Month
Jan      6200
Mar      6200
Apr      6000
Feb      5600
May      4125
Jul      3100
Aug      3100
Oct      3100
Dec      3100
Jun      3000
Sep      3000
Nov      3000
Name: count, dtype: int64
```

```
[263]: # Calculate proportions (relative frequencies)
proportions = data['Category'].value_counts(normalize=True)

# Convert to percentages
percentages = proportions * 100

# Display proportions and percentages
print("Proportions (relative frequencies):")
print(proportions)
print("\nPercentages:")
print(percentages)
```

```
Proportions (relative frequencies):
Category
Groceries      0.202766
Toys           0.200828
Clothing       0.200262
Furniture      0.199798
Electronics    0.196345
Name: proportion, dtype: float64
```

Percentages:

Category

Groceries 20.276628

Toys 20.082786

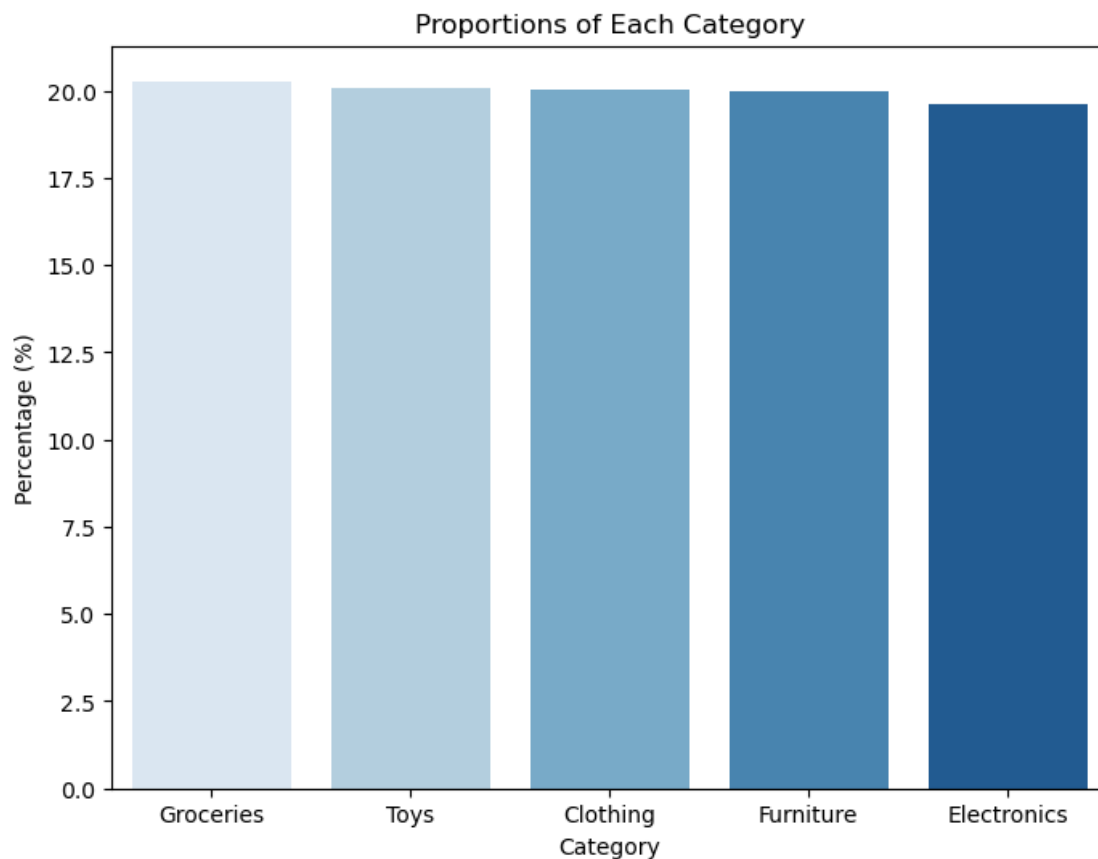
Clothing 20.026249

Furniture 19.979808

Electronics 19.634528

Name: proportion, dtype: float64

```
[265]: # Plotting a bar chart of proportions (percentages)
plt.figure(figsize=(8, 6))
sns.barplot(x=percentages.index, y=percentages.values, palette="Blues")
plt.title("Proportions of Each Category")
plt.xlabel("Category")
plt.ylabel("Percentage (%)")
plt.show()
```



```
[267]: # Create a contingency table (cross-tabulation)
contingency_table = pd.crosstab(data['Category'], data['Holiday/Promotion'])
```

```

# Perform the Chi-Square test
chi2, p, dof, expected = chi2_contingency(contingency_table)

# Display results
print(f"Contingency Table:\n{contingency_table}")
print(f"\nChi-Square Statistic: {chi2}")
print(f"P-value: {p}")
print(f"Degrees of Freedom: {dof}")
print(f"Expected Frequencies:\n{expected}")

# Interpretation
if p < 0.05:
    print("\nThere is a significant association between 'Category' and 'H/P'.")
else:
    print("\nThere is no significant association between 'Category' and 'H/p'.")

```

Contingency Table:

| Holiday/Promotion | 0    | 1    |
|-------------------|------|------|
| Category          |      |      |
| Clothing          | 4962 | 4956 |
| Electronics       | 4880 | 4844 |
| Furniture         | 4919 | 4976 |
| Groceries         | 4966 | 5076 |
| Toys              | 5033 | 4913 |

Chi-Square Statistic: 3.117508610339275

P-value: 0.5383567875978967

Degrees of Freedom: 4

Expected Frequencies:

```

[[4958.49934377 4959.50065623]
 [4861.5091368  4862.4908632 ]
 [4947.0005048  4947.9994952 ]
 [5020.4930843  5021.5069157 ]
 [4972.49793034 4973.50206966]]

```

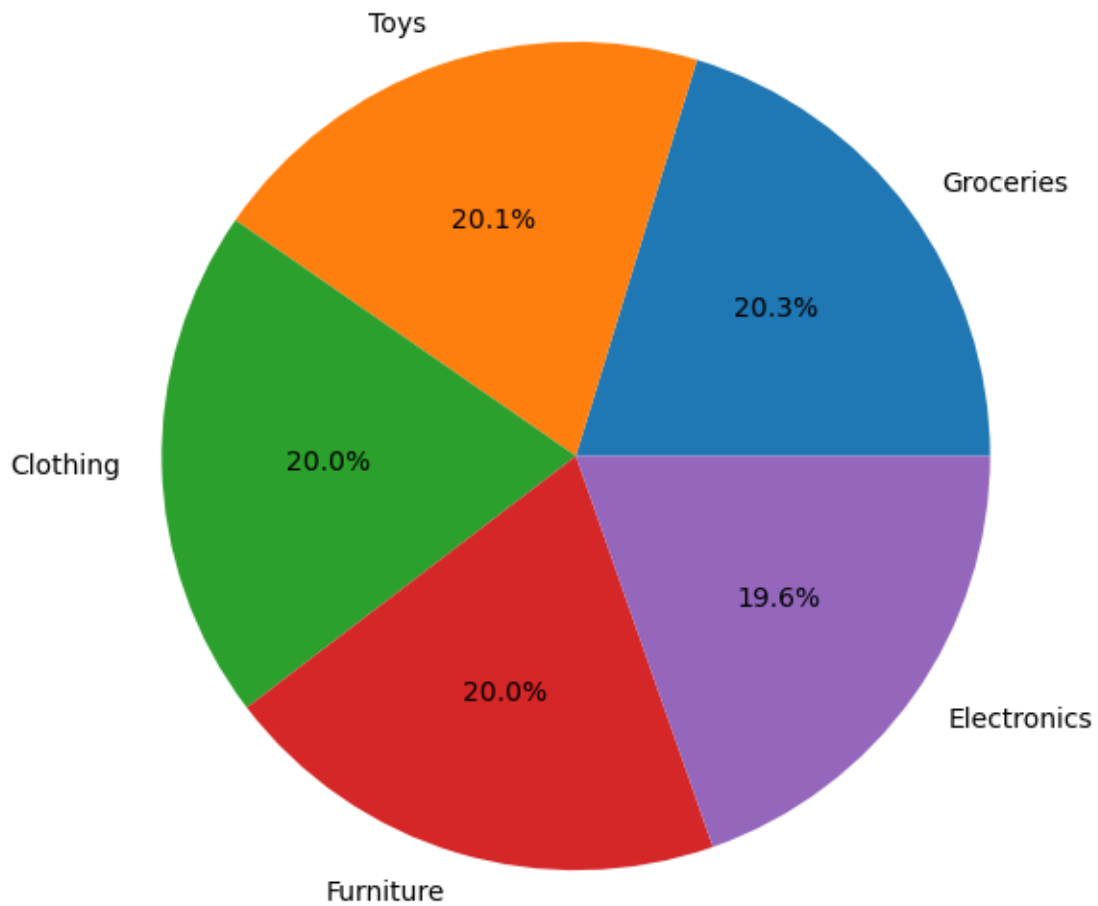
There is no significant association between 'Category' and 'H/p'.

[ ]:

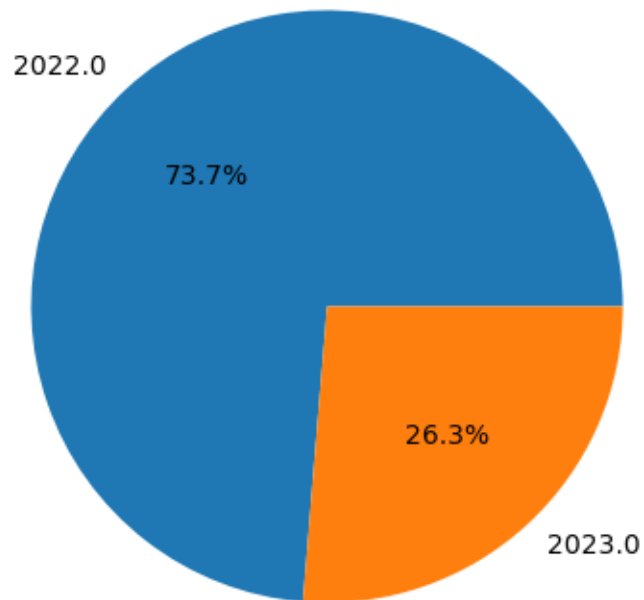
[ ]:

## 23 10. Pie-Charts Visualization

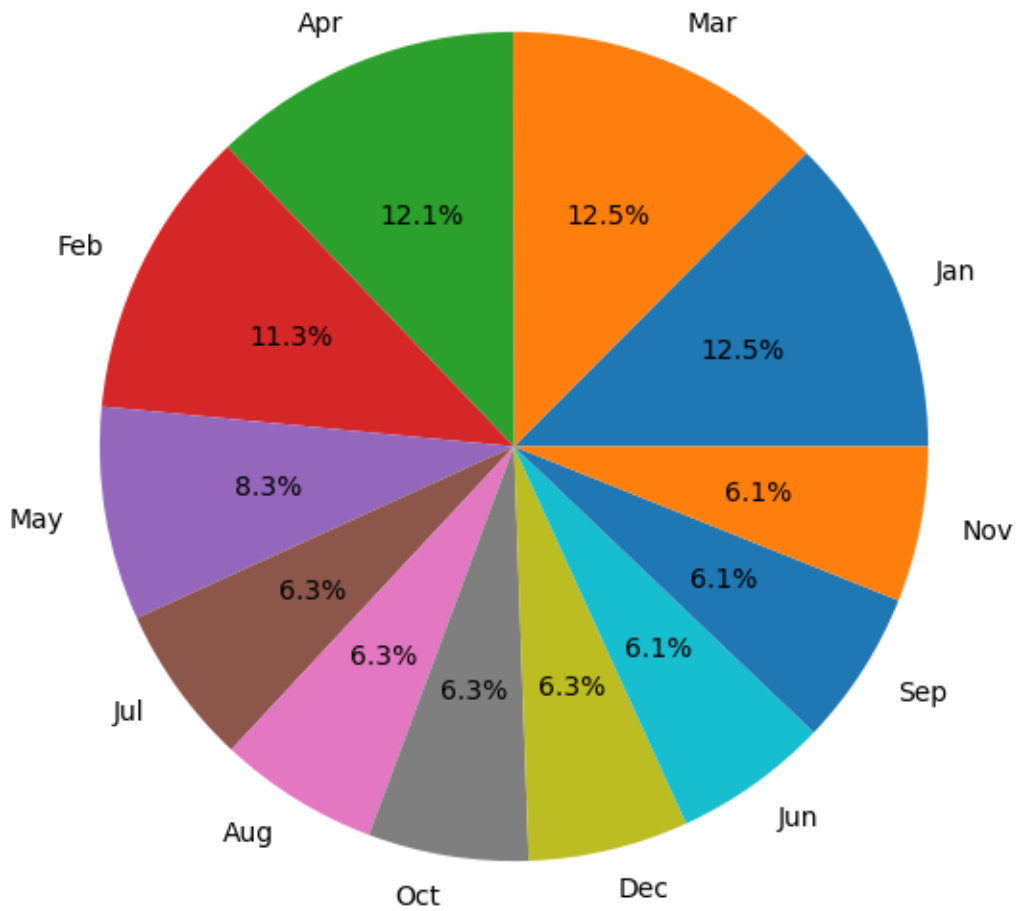
```
[269]: plt.figure(figsize=(7,7))
top_genres = data['Category'].value_counts()
plt.pie(top_genres, labels=top_genres.index, autopct='%1.1f%%')
plt.show()
```



```
[271]: plt.figure(figsize=(6,5))
top_genres = data['Year'].value_counts()
plt.pie(top_genres, labels=top_genres.index, autopct='%1.1f%%')
plt.show()
```



```
[273]: plt.figure(figsize=(7,7))
top_genres = data['Name of Month'].value_counts()
plt.pie(top_genres, labels=top_genres.index, autopct='%1.1f%%')
plt.show()
```



## 24 11. Explore through apply different method/way

```
[311]: data[data['Category'].isin(["Toys"])]
```

```
[311]:
```

|       | Date       | Store ID | Product ID | Category | Region | Inventory Level | \ |
|-------|------------|----------|------------|----------|--------|-----------------|---|
| 1     | 2022-01-01 | S001     | P0002      | Toys     | South  | 204             |   |
| 2     | 2022-01-01 | S001     | P0003      | Toys     | West   | 102             |   |
| 3     | 2022-01-01 | S001     | P0004      | Toys     | North  | 469             |   |
| 9     | 2022-01-01 | S001     | P0010      | Toys     | South  | 108             |   |
| 12    | 2022-01-01 | S001     | P0013      | Toys     | South  | 96              |   |
| ...   | ...        | ...      | ...        | ...      | ...    | ...             |   |
| 49512 | 2023-05-11 | S001     | P0013      | Toys     | South  | 272             |   |
| 49515 | 2023-05-11 | S001     | P0016      | Toys     | East   | 281             |   |

|       |            |      |       |      |       |     |
|-------|------------|------|-------|------|-------|-----|
| 49516 | 2023-05-11 | S001 | P0017 | Toys | West  | 437 |
| 49518 | 2023-05-11 | S001 | P0019 | Toys | South | 431 |
| 49521 | 2023-05-11 | S002 | P0002 | Toys | South | 443 |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 1     | 150        | 66.0          | 144.0           | 63.0  | 20       |   |
| 2     | 65         | 51.0          | 74.0            | 28.0  | 10       |   |
| 3     | 61         | 164.0         | 62.0            | 33.0  | 10       |   |
| 9     | 28         | 196.0         | 24.0            | 60.0  | 0        |   |
| 12    | 42         | 85.0          | 41.0            | 44.0  | 0        |   |
| ...   | ...        | ...           | ...             | ...   | ...      |   |
| 49512 | 52         | 135.0         | 47.0            | 33.0  | 10       |   |
| 49515 | 54         | 161.0         | 58.0            | 19.0  | 0        |   |
| 49516 | 329        | 89.0          | 340.0           | 31.0  | 5        |   |
| 49518 | 418        | 200.0         | 419.0           | 26.0  | 5        |   |
| 49521 | 77         | 53.0          | 79.0            | 33.0  | 20       |   |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality | \ |
|-------|-------------------|-------------------|--------------------|-------------|---|
| 1     | Sunny             | False             | 66                 | Autumn      |   |
| 2     | Sunny             | True              | 31                 | Summer      |   |
| 3     | Cloudy            | True              | 35                 | Autumn      |   |
| 9     | Rainy             | True              | 61                 | Winter      |   |
| 12    | Cloudy            | False             | 46                 | Spring      |   |
| ...   | ...               | ...               | ...                | ...         |   |
| 49512 | Cloudy            | True              | 30                 | Spring      |   |
| 49515 | Sunny             | True              | 16                 | Winter      |   |
| 49516 | Cloudy            | False             | 33                 | Summer      |   |
| 49518 | Cloudy            | False             | 27                 | Winter      |   |
| 49521 | Cloudy            | False             | 32                 | Winter      |   |

|       | Year | Month | Name of Month | Z-Score   |
|-------|------|-------|---------------|-----------|
| 1     | 2022 | 1     | Jan           | -0.544508 |
| 2     | 2022 | 1     | Jan           | -1.330783 |
| 3     | 2022 | 1     | Jan           | 1.498263  |
| 9     | 2022 | 1     | Jan           | -1.284531 |
| 12    | 2022 | 1     | Jan           | -1.377034 |
| ...   | ...  | ...   | ...           | ...       |
| 49512 | 2023 | 5     | May           | -0.020325 |
| 49515 | 2023 | 5     | May           | 0.049052  |
| 49516 | 2023 | 5     | May           | 1.251589  |
| 49518 | 2023 | 5     | May           | 1.205338  |
| 49521 | 2023 | 5     | May           | 1.297841  |

[9946 rows x 19 columns]

```
[313]: data[(data['Category'] == 'Furniture' ) & (data['Year'] == 2022)]
```

[313]:

|       | Date       | Store ID | Product ID | Category  | Region | Inventory Level | \ |
|-------|------------|----------|------------|-----------|--------|-----------------|---|
| 6     | 2022-01-01 | S001     | P0007      | Furniture | East   | 359             |   |
| 10    | 2022-01-01 | S001     | P0011      | Furniture | South  | 258             |   |
| 22    | 2022-01-01 | S002     | P0003      | Furniture | East   | 445             |   |
| 25    | 2022-01-01 | S002     | P0006      | Furniture | North  | 492             |   |
| 52    | 2022-01-01 | S003     | P0013      | Furniture | North  | 51              |   |
| ...   | ...        | ...      | ...        | ...       | ...    | ...             |   |
| 36482 | 2022-12-31 | S005     | P0003      | Furniture | West   | 109             |   |
| 36487 | 2022-12-31 | S005     | P0008      | Furniture | West   | 325             |   |
| 36492 | 2022-12-31 | S005     | P0013      | Furniture | North  | 256             |   |
| 36494 | 2022-12-31 | S005     | P0015      | Furniture | North  | 183             |   |
| 36499 | 2022-12-31 | S005     | P0020      | Furniture | North  | 140             |   |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 6     | 97         | 167.0         | 108.0           | 34.0  | 10       |   |
| 10    | 150        | 153.0         | 152.0           | 59.0  | 10       |   |
| 22    | 182        | 119.0         | 176.0           | 20.0  | 20       |   |
| 25    | 250        | 168.0         | 242.0           | 86.0  | 20       |   |
| 52    | 17         | 172.0         | 12.0            | 74.0  | 20       |   |
| ...   | ...        | ...           | ...             | ...   | ...      |   |
| 36482 | 8          | 52.0          | 0.0             | 33.0  | 0        |   |
| 36487 | 141        | 187.0         | 155.0           | 39.0  | 10       |   |
| 36492 | 203        | 146.0         | 220.0           | 64.0  | 5        |   |
| 36494 | 109        | 74.0          | 122.0           | 77.0  | 5        |   |
| 36499 | 41         | 43.0          | 31.0            | 55.0  | 10       |   |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality | \ |
|-------|-------------------|-------------------|--------------------|-------------|---|
| 6     | Rainy             | True              | 37                 | Winter      |   |
| 10    | Sunny             | True              | 61                 | Spring      |   |
| 22    | Snowy             | True              | 22                 | Spring      |   |
| 25    | Cloudy            | False             | 87                 | Winter      |   |
| 52    | Cloudy            | True              | 77                 | Summer      |   |
| ...   | ...               | ...               | ...                | ...         |   |
| 36482 | Snowy             | False             | 36                 | Winter      |   |
| 36487 | Snowy             | True              | 40                 | Winter      |   |
| 36492 | Cloudy            | True              | 60                 | Winter      |   |
| 36494 | Cloudy            | False             | 74                 | Summer      |   |
| 36499 | Rainy             | True              | 52                 | Autumn      |   |

|       | Year | Month | Name of Month | Z-Score   |
|-------|------|-------|---------------|-----------|
| 6     | 2022 | 1     | Jan           | 0.650320  |
| 10    | 2022 | 1     | Jan           | -0.128245 |
| 22    | 2022 | 1     | Jan           | 1.313258  |
| 25    | 2022 | 1     | Jan           | 1.675561  |
| 52    | 2022 | 1     | Jan           | -1.723920 |
| ...   | ...  | ...   | ...           | ...       |
| 36482 | 2022 | 12    | Dec           | -1.276823 |

|       |      |    |     |           |
|-------|------|----|-----|-----------|
| 36487 | 2022 | 12 | Dec | 0.388229  |
| 36492 | 2022 | 12 | Dec | -0.143663 |
| 36494 | 2022 | 12 | Dec | -0.706388 |
| 36499 | 2022 | 12 | Dec | -1.037857 |

[7290 rows x 19 columns]

```
[315]: data[(data['Category'] == 'Furniture' ) & (data['Month'] == 12)]
```

```
[315]:
```

|       | Date       | Store ID | Product ID | Category  | Region | Inventory Level | \ |
|-------|------------|----------|------------|-----------|--------|-----------------|---|
| 33405 | 2022-12-01 | S001     | P0006      | Furniture | East   | 117             |   |
| 33413 | 2022-12-01 | S001     | P0014      | Furniture | North  | 238             |   |
| 33419 | 2022-12-01 | S001     | P0020      | Furniture | South  | 451             |   |
| 33424 | 2022-12-01 | S002     | P0005      | Furniture | East   | 132             |   |
| 33428 | 2022-12-01 | S002     | P0009      | Furniture | South  | 219             |   |
| ...   | ...        | ...      | ...        | ...       | ...    | ...             |   |
| 36482 | 2022-12-31 | S005     | P0003      | Furniture | West   | 109             |   |
| 36487 | 2022-12-31 | S005     | P0008      | Furniture | West   | 325             |   |
| 36492 | 2022-12-31 | S005     | P0013      | Furniture | North  | 256             |   |
| 36494 | 2022-12-31 | S005     | P0015      | Furniture | North  | 183             |   |
| 36499 | 2022-12-31 | S005     | P0020      | Furniture | North  | 140             |   |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 33405 | 18         | 134.0         | 21.0            | 63.0  | 20       |   |
| 33413 | 108        | 177.0         | 122.0           | 11.0  | 5        |   |
| 33419 | 139        | 76.0          | 142.0           | 25.0  | 5        |   |
| 33424 | 7          | 22.0          | 0.0             | 71.0  | 5        |   |
| 33428 | 74         | 147.0         | 84.0            | 73.0  | 10       |   |
| ...   | ...        | ...           | ...             | ...   | ...      |   |
| 36482 | 8          | 52.0          | 0.0             | 33.0  | 0        |   |
| 36487 | 141        | 187.0         | 155.0           | 39.0  | 10       |   |
| 36492 | 203        | 146.0         | 220.0           | 64.0  | 5        |   |
| 36494 | 109        | 74.0          | 122.0           | 77.0  | 5        |   |
| 36499 | 41         | 43.0          | 31.0            | 55.0  | 10       |   |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality | \ |
|-------|-------------------|-------------------|--------------------|-------------|---|
| 33405 | Rainy             | False             | 62                 | Winter      |   |
| 33413 | Rainy             | False             | 13                 | Spring      |   |
| 33419 | Rainy             | False             | 20                 | Summer      |   |
| 33424 | Snowy             | False             | 67                 | Summer      |   |
| 33428 | Rainy             | True              | 70                 | Summer      |   |
| ...   | ...               | ...               | ...                | ...         |   |
| 36482 | Snowy             | False             | 36                 | Winter      |   |
| 36487 | Snowy             | True              | 40                 | Winter      |   |
| 36492 | Cloudy            | True              | 60                 | Winter      |   |
| 36494 | Cloudy            | False             | 74                 | Summer      |   |
| 36499 | Rainy             | True              | 52                 | Autumn      |   |

|       | Year | Month | Name of Month | Z-Score   |
|-------|------|-------|---------------|-----------|
| 33405 | 2022 | 12    | Dec           | -1.215154 |
| 33413 | 2022 | 12    | Dec           | -0.282417 |
| 33419 | 2022 | 12    | Dec           | 1.359509  |
| 33424 | 2022 | 12    | Dec           | -1.099526 |
| 33428 | 2022 | 12    | Dec           | -0.428880 |
| ...   | ...  | ...   | ...           | ...       |
| 36482 | 2022 | 12    | Dec           | -1.276823 |
| 36487 | 2022 | 12    | Dec           | 0.388229  |
| 36492 | 2022 | 12    | Dec           | -0.143663 |
| 36494 | 2022 | 12    | Dec           | -0.706388 |
| 36499 | 2022 | 12    | Dec           | -1.037857 |

[640 rows x 19 columns]

```
[317]: data[(data['Category'] == 'Groceries') & (data['Year'] == 2022)]
```

```
[317]:
```

|       | Date       | Store ID | Product ID | Category  | Region | Inventory Level | \ |
|-------|------------|----------|------------|-----------|--------|-----------------|---|
| 0     | 2022-01-01 | S001     | P0001      | Groceries | North  | 231             |   |
| 5     | 2022-01-01 | S001     | P0006      | Groceries | South  | 138             |   |
| 20    | 2022-01-01 | S002     | P0001      | Groceries | South  | 343             |   |
| 26    | 2022-01-01 | S002     | P0007      | Groceries | West   | 460             |   |
| 30    | 2022-01-01 | S002     | P0011      | Groceries | West   | 67              |   |
| ...   | ...        | ...      | ...        | ...       | ...    | ...             |   |
| 36488 | 2022-12-31 | S005     | P0009      | Groceries | North  | 436             |   |
| 36489 | 2022-12-31 | S005     | P0010      | Groceries | West   | 482             |   |
| 36490 | 2022-12-31 | S005     | P0011      | Groceries | South  | 377             |   |
| 36491 | 2022-12-31 | S005     | P0012      | Groceries | South  | 100             |   |
| 36496 | 2022-12-31 | S005     | P0017      | Groceries | North  | 484             |   |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 0     | 127        | 55.0          | 135.0           | 34.0  | 20       |   |
| 5     | 128        | 102.0         | 139.0           | 77.0  | 10       |   |
| 20    | 104        | 144.0         | 112.0           | 33.0  | 20       |   |
| 26    | 393        | 70.0          | 401.0           | 91.0  | 5        |   |
| 30    | 13         | 33.0          | 7.0             | 35.0  | 0        |   |
| ...   | ...        | ...           | ...             | ...   | ...      |   |
| 36488 | 147        | 25.0          | 139.0           | 17.0  | 10       |   |
| 36489 | 41         | 135.0         | 52.0            | 63.0  | 0        |   |
| 36490 | 3          | 72.0          | 8.0             | 66.0  | 10       |   |
| 36491 | 55         | 75.0          | 49.0            | 52.0  | 5        |   |
| 36496 | 341        | 195.0         | 334.0           | 48.0  | 15       |   |

|   | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality | \ |
|---|-------------------|-------------------|--------------------|-------------|---|
| 0 | Rainy             | False             | 30                 | Autumn      |   |
| 5 | Sunny             | True              | 79                 | Winter      |   |

|       |        |       |     |        |
|-------|--------|-------|-----|--------|
| 20    | Sunny  | True  | 31  | Spring |
| 26    | Sunny  | False | 93  | Summer |
| 30    | Snowy  | True  | 34  | Autumn |
| ...   | ...    | ...   | ... | ...    |
| 36488 | Rainy  | False | 15  | Spring |
| 36489 | Snowy  | False | 68  | Autumn |
| 36490 | Rainy  | True  | 68  | Summer |
| 36491 | Snowy  | True  | 52  | Winter |
| 36496 | Cloudy | False | 44  | Summer |

|       | Year | Month | Name of Month | Z-Score   |
|-------|------|-------|---------------|-----------|
| 0     | 2022 | 1     | Jan           | -0.336377 |
| 5     | 2022 | 1     | Jan           | -1.053274 |
| 20    | 2022 | 1     | Jan           | 0.526983  |
| 26    | 2022 | 1     | Jan           | 1.428886  |
| 30    | 2022 | 1     | Jan           | -1.600583 |
| ...   | ...  | ...   | ...           | ...       |
| 36488 | 2022 | 12    | Dec           | 1.243881  |
| 36489 | 2022 | 12    | Dec           | 1.598475  |
| 36490 | 2022 | 12    | Dec           | 0.789075  |
| 36491 | 2022 | 12    | Dec           | -1.346200 |
| 36496 | 2022 | 12    | Dec           | 1.613892  |

[7357 rows x 19 columns]

```
[319]: data[(data['Region'] == 'South' ) & (data['Year'] == 2022) & (data['Discount']_
↳== 20) & (data['Category'] == "Toys") & (data['Month'] == 10) &_
↳(data['Seasonality'] == 'Spring') & (data['Weather Condition'] == "Sunny")]
```

```
[319]:
```

|       | Date       | Store ID | Product ID | Category | Region | Inventory Level | \ |
|-------|------------|----------|------------|----------|--------|-----------------|---|
| 27393 | 2022-10-01 | S005     | P0014      | Toys     | South  | 78              |   |
| 27708 | 2022-10-05 | S001     | P0009      | Toys     | South  | 83              |   |
| 27990 | 2022-10-07 | S005     | P0011      | Toys     | South  | 346             |   |
| 30295 | 2022-10-30 | S005     | P0016      | Toys     | South  | 70              |   |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 27393 | 63         | 200.0         | 58.0            | 81.0  | 20       |   |
| 27708 | 37         | 54.0          | 30.0            | 73.0  | 20       |   |
| 27990 | 169        | 26.0          | 163.0           | 92.0  | 20       |   |
| 30295 | 1          | 141.0         | 6.0             | 72.0  | 20       |   |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality | \ |
|-------|-------------------|-------------------|--------------------|-------------|---|
| 27393 | Sunny             | True              | 84                 | Spring      |   |
| 27708 | Sunny             | False             | 73                 | Spring      |   |
| 27990 | Sunny             | True              | 93                 | Spring      |   |
| 30295 | Sunny             | False             | 77                 | Spring      |   |

|       | Year | Month | Name of Month | Z-Score   |
|-------|------|-------|---------------|-----------|
| 27393 | 2022 | 10    | Oct           | -1.515789 |
| 27708 | 2022 | 10    | Oct           | -1.477246 |
| 27990 | 2022 | 10    | Oct           | 0.550109  |
| 30295 | 2022 | 10    | Oct           | -1.577457 |

```
[321]: data[(data['Region'] == 'North' ) & (data['Year'] == 2022) & (data['Discount']_
↪== 10) & (data['Category'] == "Clothing") & (data['Month'] == 10) &_
↪(data['Seasonality'] == 'Winter') & (data['Weather Condition'] == "Snowy")]
```

```
[321]:
```

|       | Date       | Store ID | Product ID | Category | Region | Inventory Level | \ |
|-------|------------|----------|------------|----------|--------|-----------------|---|
| 28472 | 2022-10-12 | S004     | P0013      | Clothing | North  | 487             |   |
| 28790 | 2022-10-15 | S005     | P0011      | Clothing | North  | 477             |   |
| 29049 | 2022-10-18 | S003     | P0010      | Clothing | North  | 412             |   |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 28472 | 276        | 53.0          | 283.0           | 87.0  | 10       |   |
| 28790 | 48         | 79.0          | 38.0            | 55.0  | 10       |   |
| 29049 | 236        | 57.0          | 250.0           | 22.0  | 10       |   |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality | \ |
|-------|-------------------|-------------------|--------------------|-------------|---|
| 28472 | Snowy             | False             | 85                 | Winter      |   |
| 28790 | Snowy             | True              | 57                 | Winter      |   |
| 29049 | Snowy             | True              | 24                 | Winter      |   |

|       | Year | Month | Name of Month | Z-Score  |
|-------|------|-------|---------------|----------|
| 28472 | 2022 | 10    | Oct           | 1.637018 |
| 28790 | 2022 | 10    | Oct           | 1.559932 |
| 29049 | 2022 | 10    | Oct           | 1.058875 |

```
[323]: data[(data['Region'] == 'East' ) & (data['Year'] == 2022) & (data['Discount']_
↪== 20) & (data['Category'] == "Clothing") & (data['Month'] == 8) &_
↪(data['Seasonality'] == 'Winter') & (data['Weather Condition'] == "Snowy")]
```

```
[323]:
```

|       | Date       | Store ID | Product ID | Category | Region | Inventory Level | \ |
|-------|------------|----------|------------|----------|--------|-----------------|---|
| 21759 | 2022-08-06 | S003     | P0020      | Clothing | East   | 146             |   |
| 21972 | 2022-08-08 | S004     | P0013      | Clothing | East   | 170             |   |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 21759 | 51         | 171.0         | 47.0            | 100.0 | 20       |   |
| 21972 | 69         | 170.0         | 81.0            | 16.0  | 20       |   |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality | \ |
|-------|-------------------|-------------------|--------------------|-------------|---|
| 21759 | Snowy             | False             | 104                | Winter      |   |
| 21972 | Snowy             | False             | 13                 | Winter      |   |

|  | Year | Month | Name of Month | Z-Score |
|--|------|-------|---------------|---------|
|--|------|-------|---------------|---------|

|       |      |   |     |           |
|-------|------|---|-----|-----------|
| 21759 | 2022 | 8 | Aug | -0.991606 |
| 21972 | 2022 | 8 | Aug | -0.806600 |

```
[325]: data[(data['Category'] == 'Electronics' ) & (data['Year'] == 2023) &
↳ ((data['Region'] == "West") | (data['Name of Month'] == 'June'))]
```

```
[325]:
```

|       | Date       | Store ID | Product ID | Category    | Region | Inventory Level | \ |
|-------|------------|----------|------------|-------------|--------|-----------------|---|
| 36509 | 2023-01-01 | S001     | P0010      | Electronics | West   | 233             |   |
| 36511 | 2023-01-01 | S001     | P0012      | Electronics | West   | 373             |   |
| 36514 | 2023-01-01 | S001     | P0015      | Electronics | West   | 365             |   |
| 36537 | 2023-01-01 | S002     | P0018      | Electronics | West   | 128             |   |
| 36545 | 2023-01-01 | S003     | P0006      | Electronics | West   | 92              |   |
| ...   | ...        | ...      | ...        | ...         | ...    | ...             |   |
| 49466 | 2023-05-10 | S004     | P0007      | Electronics | West   | 148             |   |
| 49476 | 2023-05-10 | S004     | P0017      | Electronics | West   | 125             |   |
| 49510 | 2023-05-11 | S001     | P0011      | Electronics | West   | 248             |   |
| 49517 | 2023-05-11 | S001     | P0018      | Electronics | West   | 247             |   |
| 49519 | 2023-05-11 | S001     | P0020      | Electronics | West   | 360             |   |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount | \ |
|-------|------------|---------------|-----------------|-------|----------|---|
| 36509 | 150        | 190.0         | 165.0           | 13.0  | 0        |   |
| 36511 | 3          | 198.0         | 18.0            | 21.0  | 20       |   |
| 36514 | 312        | 200.0         | 308.0           | 41.0  | 20       |   |
| 36537 | 102        | 69.0          | 105.0           | 30.0  | 20       |   |
| 36545 | 52         | 95.0          | 61.0            | 34.0  | 5        |   |
| ...   | ...        | ...           | ...             | ...   | ...      |   |
| 49466 | 129        | 22.0          | 148.0           | 31.0  | 10       |   |
| 49476 | 8          | 182.0         | 0.0             | 18.0  | 20       |   |
| 49510 | 54         | 120.0         | 68.0            | 73.0  | 0        |   |
| 49517 | 69         | 107.0         | 87.0            | 11.0  | 20       |   |
| 49519 | 159        | 117.0         | 158.0           | 92.0  | 20       |   |

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality | \ |
|-------|-------------------|-------------------|--------------------|-------------|---|
| 36509 | Cloudy            | False             | 16                 | Summer      |   |
| 36511 | Rainy             | True              | 22                 | Summer      |   |
| 36514 | Cloudy            | True              | 42                 | Spring      |   |
| 36537 | Sunny             | True              | 34                 | Spring      |   |
| 36545 | Snowy             | False             | 29                 | Winter      |   |
| ...   | ...               | ...               | ...                | ...         |   |
| 49466 | Sunny             | False             | 26                 | Winter      |   |
| 49476 | Sunny             | True              | 17                 | Spring      |   |
| 49510 | Sunny             | True              | 71                 | Summer      |   |
| 49517 | Snowy             | False             | 11                 | Summer      |   |
| 49519 | Sunny             | False             | 89                 | Autumn      |   |

|       | Year | Month | Name of Month | Z-Score   |
|-------|------|-------|---------------|-----------|
| 36509 | 2023 | 1     | Jan           | -0.320960 |

|       |      |     |     |           |
|-------|------|-----|-----|-----------|
| 36511 | 2023 | 1   | Jan | 0.758240  |
| 36514 | 2023 | 1   | Jan | 0.696572  |
| 36537 | 2023 | 1   | Jan | -1.130360 |
| 36545 | 2023 | 1   | Jan | -1.407869 |
| ...   | ...  | ... | ... | ...       |
| 49466 | 2023 | 5   | May | -0.976188 |
| 49476 | 2023 | 5   | May | -1.153486 |
| 49510 | 2023 | 5   | May | -0.205331 |
| 49517 | 2023 | 5   | May | -0.213040 |
| 49519 | 2023 | 5   | May | 0.658029  |

[631 rows x 19 columns]

[327]: *# Replace 1 with True and 0 with False*

```
data['Holiday/Promotion'] = data['Holiday/Promotion'].replace({1: True, 0:
↪False})
```

[329]:

```
by = data.sort_values("Category", axis=0, inplace=False, ascending=True)
by
be = by.reset_index()
be.drop(columns='index')
```

[329]:

|       | Date       | Store ID | Product ID | Category | Region | Inventory Level \ |
|-------|------------|----------|------------|----------|--------|-------------------|
| 0     | 2022-08-16 | S002     | P0001      | Clothing | North  | 166               |
| 1     | 2022-03-26 | S003     | P0010      | Clothing | North  | 317               |
| 2     | 2022-10-19 | S002     | P0005      | Clothing | East   | 185               |
| 3     | 2022-03-26 | S003     | P0012      | Clothing | East   | 54                |
| 4     | 2022-03-26 | S003     | P0014      | Clothing | West   | 453               |
| ...   | ...        | ...      | ...        | ...      | ...    | ...               |
| 49520 | 2022-07-11 | S004     | P0008      | Toys     | South  | 201               |
| 49521 | 2023-02-02 | S005     | P0018      | Toys     | East   | 466               |
| 49522 | 2022-07-11 | S003     | P0005      | Toys     | East   | 445               |
| 49523 | 2022-07-12 | S004     | P0001      | Toys     | East   | 382               |
| 49524 | 2022-09-05 | S004     | P0003      | Toys     | West   | 373               |

|       | Units Sold | Units Ordered | Demand Forecast | Price | Discount \ |
|-------|------------|---------------|-----------------|-------|------------|
| 0     | 35         | 196.0         | 52.0            | 81.0  | 20         |
| 1     | 141        | 119.0         | 137.0           | 35.0  | 5          |
| 2     | 87         | 30.0          | 98.0            | 42.0  | 10         |
| 3     | 18         | 125.0         | 29.0            | 71.0  | 5          |
| 4     | 416        | 95.0          | 426.0           | 62.0  | 0          |
| ...   | ...        | ...           | ...             | ...   | ...        |
| 49520 | 174        | 33.0          | 189.0           | 74.0  | 0          |
| 49521 | 138        | 111.0         | 142.0           | 74.0  | 0          |
| 49522 | 184        | 170.0         | 200.0           | 57.0  | 20         |
| 49523 | 259        | 29.0          | 255.0           | 40.0  | 15         |

|       |    |       |      |      |    |
|-------|----|-------|------|------|----|
| 49524 | 22 | 144.0 | 26.0 | 40.0 | 10 |
|-------|----|-------|------|------|----|

|       | Weather Condition | Holiday/Promotion | Competitor Pricing | Seasonality \ |
|-------|-------------------|-------------------|--------------------|---------------|
| 0     | Snowy             | True              | 79                 | Autumn        |
| 1     | Sunny             | True              | 36                 | Spring        |
| 2     | Rainy             | True              | 46                 | Summer        |
| 3     | Snowy             | False             | 67                 | Summer        |
| 4     | Snowy             | True              | 57                 | Summer        |
| ...   | ...               | ...               | ...                | ...           |
| 49520 | Rainy             | False             | 78                 | Summer        |
| 49521 | Cloudy            | True              | 74                 | Summer        |
| 49522 | Rainy             | True              | 55                 | Winter        |
| 49523 | Cloudy            | True              | 36                 | Summer        |
| 49524 | Sunny             | False             | 41                 | Autumn        |

|       | Year | Month | Name of Month | Z-Score   |
|-------|------|-------|---------------|-----------|
| 0     | 2022 | 8     | Aug           | -0.837434 |
| 1     | 2022 | 3     | Mar           | 0.326560  |
| 2     | 2022 | 10    | Oct           | -0.690971 |
| 3     | 2022 | 3     | Mar           | -1.700794 |
| 4     | 2022 | 3     | Mar           | 1.374926  |
| ...   | ...  | ...   | ...           | ...       |
| 49520 | 2022 | 7     | Jul           | -0.567634 |
| 49521 | 2023 | 2     | Feb           | 1.475138  |
| 49522 | 2022 | 7     | Jul           | 1.313258  |
| 49523 | 2022 | 7     | Jul           | 0.827618  |
| 49524 | 2022 | 9     | Sep           | 0.758240  |

[49525 rows x 19 columns]

## 25 Pivot\_Table, Aggergate and Groupby with Columns, etc..

```
[355]: pivot = data.pivot_table(index='Date', values='Price', aggfunc=['sum', 'mean', 'median'],
    ↪ 'median'])
pivot
```

```
[355]:
```

|            | sum<br>Price | mean<br>Price | median<br>Price |
|------------|--------------|---------------|-----------------|
| Date       |              |               |                 |
| 2022-01-01 | 5754.0       | 57.54         | 59.5            |
| 2022-01-02 | 6064.0       | 60.64         | 63.5            |
| 2022-01-03 | 5681.0       | 56.81         | 58.0            |
| 2022-01-04 | 5296.0       | 52.96         | 53.0            |
| 2022-01-05 | 5600.0       | 56.00         | 54.0            |
| ...        | ...          | ...           | ...             |
| 2023-05-07 | 5537.0       | 55.37         | 57.0            |

|            |        |       |      |
|------------|--------|-------|------|
| 2023-05-08 | 5114.0 | 51.14 | 48.0 |
| 2023-05-09 | 5521.0 | 55.21 | 55.0 |
| 2023-05-10 | 5341.0 | 53.41 | 55.5 |
| 2023-05-11 | 1231.0 | 49.24 | 44.0 |

[496 rows x 3 columns]

```
[357]: pivot.head(20)
```

```
[357]:
```

|            | sum    | mean  | median |
|------------|--------|-------|--------|
|            | Price  | Price | Price  |
| Date       |        |       |        |
| 2022-01-01 | 5754.0 | 57.54 | 59.5   |
| 2022-01-02 | 6064.0 | 60.64 | 63.5   |
| 2022-01-03 | 5681.0 | 56.81 | 58.0   |
| 2022-01-04 | 5296.0 | 52.96 | 53.0   |
| 2022-01-05 | 5600.0 | 56.00 | 54.0   |
| 2022-01-06 | 5698.0 | 56.98 | 57.0   |
| 2022-01-07 | 5871.0 | 58.71 | 61.0   |
| 2022-01-08 | 5341.0 | 53.41 | 54.0   |
| 2022-01-09 | 5863.0 | 58.63 | 57.5   |
| 2022-01-10 | 5487.0 | 54.87 | 56.5   |
| 2022-01-11 | 5496.0 | 54.96 | 54.0   |
| 2022-01-12 | 5835.0 | 58.35 | 59.0   |
| 2022-01-13 | 5442.0 | 54.42 | 52.5   |
| 2022-01-14 | 5683.0 | 56.83 | 59.0   |
| 2022-01-15 | 5444.0 | 54.44 | 52.0   |
| 2022-01-16 | 5376.0 | 53.76 | 55.0   |
| 2022-01-17 | 5559.0 | 55.59 | 58.5   |
| 2022-01-18 | 5738.0 | 57.38 | 61.0   |
| 2022-01-19 | 5284.0 | 52.84 | 51.5   |
| 2022-01-20 | 5709.0 | 57.09 | 60.0   |

```
[359]: pivot.tail(20)
```

```
[359]:
```

|            | sum    | mean  | median |
|------------|--------|-------|--------|
|            | Price  | Price | Price  |
| Date       |        |       |        |
| 2023-04-22 | 5488.0 | 54.88 | 54.5   |
| 2023-04-23 | 5118.0 | 51.18 | 47.0   |
| 2023-04-24 | 5447.0 | 54.47 | 54.5   |
| 2023-04-25 | 5523.0 | 55.23 | 54.5   |
| 2023-04-26 | 5365.0 | 53.65 | 52.5   |
| 2023-04-27 | 5325.0 | 53.25 | 54.5   |
| 2023-04-28 | 5228.0 | 52.28 | 48.0   |
| 2023-04-29 | 5768.0 | 57.68 | 61.0   |
| 2023-04-30 | 6131.0 | 61.31 | 64.5   |

|            |        |       |      |
|------------|--------|-------|------|
| 2023-05-01 | 5766.0 | 57.66 | 59.0 |
| 2023-05-02 | 5209.0 | 52.09 | 50.5 |
| 2023-05-03 | 5696.0 | 56.96 | 58.5 |
| 2023-05-04 | 5379.0 | 53.79 | 55.0 |
| 2023-05-05 | 5300.0 | 53.00 | 49.0 |
| 2023-05-06 | 6026.0 | 60.26 | 66.0 |
| 2023-05-07 | 5537.0 | 55.37 | 57.0 |
| 2023-05-08 | 5114.0 | 51.14 | 48.0 |
| 2023-05-09 | 5521.0 | 55.21 | 55.0 |
| 2023-05-10 | 5341.0 | 53.41 | 55.5 |
| 2023-05-11 | 1231.0 | 49.24 | 44.0 |

```
[391]: data.groupby(['Category']).agg({'Price':'sum', 'Discount':'sum', 'Month':'sum'})
```

```
[391]:
```

|             | Price    | Discount | Month |
|-------------|----------|----------|-------|
| Category    |          |          |       |
| Clothing    | 544038.0 | 99255    | 54900 |
| Electronics | 540548.0 | 97915    | 54076 |
| Furniture   | 549278.0 | 99710    | 54524 |
| Groceries   | 557443.0 | 99790    | 54939 |
| Toys        | 549033.0 | 99660    | 54886 |

```
[393]: data.groupby(['Region', 'Category']).value_counts()
```

```
[393]:
```

| Region            | Category   | Date       | Store ID    | Product ID | Inventory Level | Units Sold    |
|-------------------|------------|------------|-------------|------------|-----------------|---------------|
| Units Ordered     | Demand     | Forecast   | Price       | Discount   | Weather         | Condition     |
| Holiday/Promotion | Competitor | Pricing    | Seasonality | Year       | Month           | Name of Month |
| Z-Score           |            |            |             |            |                 |               |
| East              | Clothing   | 2022-01-01 | S001        | P0019      | 352             | 257           |
| 186.0             |            | 267.0      | 73.0        | 10         | Cloudy          | False         |
| 77                |            | Winter     | 2022        | 1          | Jan             | 0.596360 1    |
|                   |            |            | S002        | P0013      | 389             | 283           |
| 177.0             |            | 299.0      | 58.0        | 20         | Cloudy          | False         |
| 58                |            | Spring     | 2022        | 1          | Jan             | 0.881578 1    |
|                   |            | 2023-02-05 | S001        | P0005      | 448             | 273           |
| 109.0             |            | 270.0      | 25.0        | 0          | Sunny           | True          |
| 27                |            | Summer     | 2023        | 2          | Feb             | 1.336383 1    |
|                   |            | 2023-02-09 | S003        | P0007      | 206             | 40            |
| 52.0              |            | 35.0       | 78.0        | 0          | Rainy           | True          |
| 74                |            | Summer     | 2023        | 2          | Feb             | -0.529091 1   |
|                   |            | 2023-02-10 | S001        | P0004      | 481             | 226           |
| 195.0             |            | 229.0      | 84.0        | 5          | Snowy           | True          |
| 84                |            | Winter     | 2023        | 2          | Feb             | 1.590766 1    |
| ..                |            |            |             |            |                 |               |
| West              | Toys       | 2022-03-22 | S004        | P0002      | 367             | 276           |
| 81.0              |            | 268.0      | 30.0        | 0          | Sunny           | True          |
| 29                |            | Autumn     | 2022        | 3          | Mar             | 0.711989 1    |

|      |        |      |    |       |           |     |
|------|--------|------|----|-------|-----------|-----|
| 66.0 | 354.0  | 13.0 | 0  | P0008 | 479       | 361 |
| 14   | Summer | 2022 | 3  | Mar   | 1.575349  | 1   |
| 86.0 | 53.0   | 87.0 | 15 | P0012 | 58        | 38  |
| 85   | Winter | 2022 | 3  | Mar   | -1.669960 | 1   |
| 92.0 | 299.0  | 63.0 | 20 | P0007 | 356       | 309 |
| 65   | Spring | 2022 | 3  | Mar   | 0.627195  | 1   |
| 89.0 | 340.0  | 31.0 | 5  | P0017 | 437       | 329 |
| 33   | Summer | 2023 | 5  | May   | 1.251589  | 1   |

Name: count, Length: 49525, dtype: int64

```
[404]: data.pivot_table(index='Date', values='Price', aggfunc=['sum', 'mean', 'median', 'count'])
```

```
[404]:
```

|            | sum    | mean  | median | count |
|------------|--------|-------|--------|-------|
| Date       | Price  | Price | Price  | Price |
| 2022-01-01 | 5754.0 | 57.54 | 59.5   | 100   |
| 2022-01-02 | 6064.0 | 60.64 | 63.5   | 100   |
| 2022-01-03 | 5681.0 | 56.81 | 58.0   | 100   |
| 2022-01-04 | 5296.0 | 52.96 | 53.0   | 100   |
| 2022-01-05 | 5600.0 | 56.00 | 54.0   | 100   |
| ...        | ...    | ...   | ...    | ...   |
| 2023-05-07 | 5537.0 | 55.37 | 57.0   | 100   |
| 2023-05-08 | 5114.0 | 51.14 | 48.0   | 100   |
| 2023-05-09 | 5521.0 | 55.21 | 55.0   | 100   |
| 2023-05-10 | 5341.0 | 53.41 | 55.5   | 100   |
| 2023-05-11 | 1231.0 | 49.24 | 44.0   | 25    |

[496 rows x 4 columns]

```
[412]: pivot_ta = pd.pivot_table(data, values=['Inventory Level', 'Price', "Year"], index='Category')
print(pivot_ta)
```

|             | Inventory Level | Price     | Year        |
|-------------|-----------------|-----------|-------------|
| Category    |                 |           |             |
| Clothing    | 274.065437      | 54.853600 | 2022.264569 |
| Electronics | 272.709893      | 55.589058 | 2022.253908 |
| Furniture   | 276.434563      | 55.510662 | 2022.263264 |
| Groceries   | 276.029875      | 55.511153 | 2022.267377 |
| Toys        | 273.895033      | 55.201387 | 2022.265634 |

```
[416]: pivot_tab = pd.pivot_table(data, values='Competitor Pricing',
    ↪index=['Category', 'Region'])
print(pivot_tab)
```

| Competitor Pricing |        |           |
|--------------------|--------|-----------|
| Category           | Region |           |
| Clothing           | East   | 54.846515 |
|                    | North  | 55.563922 |
|                    | South  | 54.797078 |
|                    | West   | 54.449054 |
| Electronics        | East   | 55.472594 |
|                    | North  | 55.090116 |
|                    | South  | 55.939724 |
|                    | West   | 55.864854 |
| Furniture          | East   | 54.875572 |
|                    | North  | 55.831886 |
|                    | South  | 55.430624 |
|                    | West   | 55.943299 |
| Groceries          | East   | 55.798180 |
|                    | North  | 55.237961 |
|                    | South  | 56.029020 |
|                    | West   | 55.048928 |
| Toys               | East   | 55.624898 |
|                    | North  | 55.740741 |
|                    | South  | 54.581330 |
|                    | West   | 54.756680 |

```
[422]: pivot_table = pd.pivot_table(data, values=['Units Ordered', 'Units Sold'],
    ↪index='Category', columns='Region', aggfunc='sum')
print(pivot_table)
```

|             | Units Ordered |          |          |          | Units Sold |        |   |
|-------------|---------------|----------|----------|----------|------------|--------|---|
| Region      | East          | North    | South    | West     | East       | North  | \ |
| Category    |               |          |          |          |            |        |   |
| Clothing    | 284490.0      | 262601.0 | 275471.0 | 271597.0 | 342686     | 325630 |   |
| Electronics | 269771.0      | 263784.0 | 259727.0 | 273993.0 | 330654     | 328007 |   |
| Furniture   | 262095.0      | 279598.0 | 268218.0 | 278609.0 | 333433     | 353490 |   |
| Groceries   | 282823.0      | 270442.0 | 282435.0 | 272828.0 | 350887     | 336934 |   |
| Toys        | 265772.0      | 280915.0 | 272008.0 | 272857.0 | 334808     | 349392 |   |

| Region      | South  | West   |
|-------------|--------|--------|
| Category    |        |        |
| Clothing    | 339222 | 341245 |
| Electronics | 320101 | 327142 |
| Furniture   | 337284 | 338505 |
| Groceries   | 349018 | 340610 |

|      |        |        |
|------|--------|--------|
| Toys | 336945 | 338685 |
|------|--------|--------|

### 25.0.1 Basic Pivot Table

```
[430]: pivot = pd.pivot_table(data, values='Units Sold', index='Region', aggfunc='sum')
print(pivot)
```

|        | Units Sold |
|--------|------------|
| Region |            |
| East   | 1692468    |
| North  | 1693453    |
| South  | 1682570    |
| West   | 1686187    |

```
[432]: pivot = pd.pivot_table(data, values='Units Sold', index='Category',
    ↪aggfunc='sum')
print(pivot)
```

|             | Units Sold |
|-------------|------------|
| Category    |            |
| Clothing    | 1348783    |
| Electronics | 1305904    |
| Furniture   | 1362712    |
| Groceries   | 1377449    |
| Toys        | 1359830    |

### 25.0.2 Pivot Table with Multiple Aggregations

```
[426]: pivot = pd.pivot_table(data, values=['Price', 'Discount'], index='Region',
    ↪aggfunc='sum')
print(pivot)
```

|        | Discount | Price    |
|--------|----------|----------|
| Region |          |          |
| East   | 125145   | 685640.0 |
| North  | 123520   | 687289.0 |
| South  | 123270   | 682934.0 |
| West   | 124395   | 684477.0 |

### 25.0.3 Using Columns for Additional Grouping

```
[428]: pivot = pd.pivot_table(data, values='Price', index='Region',
    ↪columns='Category', aggfunc='sum', fill_value=0)
print(pivot)
```

| Category | Clothing | Electronics | Furniture | Groceries | Toys   |
|----------|----------|-------------|-----------|-----------|--------|
| Region   |          |             |           |           |        |
| East     | 139977   | 136799      | 131886    | 141000    | 135978 |

|       |        |        |        |        |        |
|-------|--------|--------|--------|--------|--------|
| North | 134330 | 132544 | 141368 | 137586 | 141461 |
| South | 134948 | 133751 | 135059 | 142676 | 136500 |
| West  | 134783 | 137454 | 140965 | 136181 | 135094 |

#### 25.0.4 Applying Multiple Aggregation Functions

```
[440]: pivot = pd.pivot_table(data, values='Demand Forecast', index='Region',
    ↪aggfunc=['sum', 'mean', 'count'])
print(pivot)
```

|                 | sum             | mean            | count           |
|-----------------|-----------------|-----------------|-----------------|
| Demand Forecast | Demand Forecast | Demand Forecast | Demand Forecast |
| Region          |                 |                 |                 |
| East            | 1743452.0       | 140.725805      | 12389           |
| North           | 1747506.0       | 141.053031      | 12389           |
| South           | 1736528.0       | 140.780543      | 12335           |
| West            | 1740490.0       | 140.226394      | 12412           |

#### 25.0.5 Adding Margins (Grand Totals)

```
[443]: pivot = pd.pivot_table(data, values=['Competitor Pricing', 'Z-Score'],
    ↪index='Category', columns='Date', aggfunc='sum', margins=True, fill_value=0)
print(pivot)
```

|             | Competitor Pricing  |                     |                     | \ |
|-------------|---------------------|---------------------|---------------------|---|
| Date        | 2022-01-01 00:00:00 | 2022-01-02 00:00:00 | 2022-01-03 00:00:00 |   |
| Category    |                     |                     |                     |   |
| Clothing    | 1267                | 1019                | 1529                |   |
| Electronics | 1249                | 1122                | 1177                |   |
| Furniture   | 698                 | 1227                | 647                 |   |
| Groceries   | 1440                | 1141                | 1369                |   |
| Toys        | 1161                | 1517                | 968                 |   |
| All         | 5815                | 6026                | 5690                |   |

|             |                     |                     |                     | \ |
|-------------|---------------------|---------------------|---------------------|---|
| Date        | 2022-01-04 00:00:00 | 2022-01-05 00:00:00 | 2022-01-06 00:00:00 |   |
| Category    |                     |                     |                     |   |
| Clothing    | 963                 | 590                 | 1197                |   |
| Electronics | 1027                | 1312                | 956                 |   |
| Furniture   | 706                 | 1060                | 1726                |   |
| Groceries   | 1056                | 1089                | 728                 |   |
| Toys        | 1537                | 1548                | 1067                |   |
| All         | 5289                | 5599                | 5674                |   |

|          |                     |                     |                     | \ |
|----------|---------------------|---------------------|---------------------|---|
| Date     | 2022-01-07 00:00:00 | 2022-01-08 00:00:00 | 2022-01-09 00:00:00 |   |
| Category |                     |                     |                     |   |
| Clothing | 1111                | 769                 | 1401                |   |

|             |      |      |      |
|-------------|------|------|------|
| Electronics | 1029 | 991  | 1685 |
| Furniture   | 1200 | 1287 | 594  |
| Groceries   | 1277 | 1298 | 666  |
| Toys        | 1240 | 992  | 1495 |
| All         | 5857 | 5337 | 5841 |

|             |                     |     |                     |                     |
|-------------|---------------------|-----|---------------------|---------------------|
|             |                     | ... | Z-Score             |                     |
| Date        | 2022-01-10 00:00:00 | ... | 2023-05-03 00:00:00 | 2023-05-04 00:00:00 |
| Category    |                     | ... |                     |                     |
| Clothing    | 1332                | ... | -0.843066           | -1.934191           |
| Electronics | 997                 | ... | 4.280304            | 2.160447            |
| Furniture   | 1092                | ... | 1.821962            | 2.163247            |
| Groceries   | 1053                | ... | 5.648214            | -2.955246           |
| Toys        | 1002                | ... | -5.269926           | -1.212416           |
| All         | 5476                | ... | 5.637489            | -1.778159           |

|             |                     |                     |                     |
|-------------|---------------------|---------------------|---------------------|
| Date        | 2023-05-05 00:00:00 | 2023-05-06 00:00:00 | 2023-05-07 00:00:00 |
| Category    |                     |                     |                     |
| Clothing    | -4.321740           | -3.157053           | -4.527795           |
| Electronics | 0.550770            | -2.079992           | -0.489887           |
| Furniture   | -7.234196           | 1.977549            | 7.210254            |
| Groceries   | 1.632017            | -2.560001           | -7.965818           |
| Toys        | -0.398800           | 2.969847            | -6.241898           |
| All         | -9.771949           | -2.849651           | -12.015144          |

|             |                     |                     |                     |
|-------------|---------------------|---------------------|---------------------|
| Date        | 2023-05-08 00:00:00 | 2023-05-09 00:00:00 | 2023-05-10 00:00:00 |
| Category    |                     |                     |                     |
| Clothing    | -1.511635           | 3.278190            | 2.009768            |
| Electronics | -5.219458           | 4.431675            | -5.136772           |
| Furniture   | -2.198390           | -0.944724           | -10.437470          |
| Groceries   | -2.037233           | -1.306303           | -0.564172           |
| Toys        | 8.911049            | -1.378481           | 2.128920            |
| All         | -2.055668           | 4.080357            | -11.999727          |

|             |                     |               |
|-------------|---------------------|---------------|
| Date        | 2023-05-11 00:00:00 | All           |
| Category    |                     |               |
| Clothing    | -2.221485           | -4.367737e+01 |
| Electronics | 0.371396            | -1.444321e+02 |
| Furniture   | -0.392445           | 1.371322e+02  |
| Groceries   | -0.240381           | 1.078427e+02  |
| Toys        | 3.971300            | -5.686542e+01 |
| All         | 1.488385            | -4.121148e-12 |

[6 rows x 994 columns]

```
[445]: pivot = pd.pivot_table(data, values='Competitor Pricing', index='Date',
    ↪columns='Category', aggfunc='sum', margins=True, fill_value=0)
print(pivot)
```

| Category            | Clothing | Electronics | Furniture | Groceries | Toys \ |
|---------------------|----------|-------------|-----------|-----------|--------|
| Date                |          |             |           |           |        |
| 2022-01-01 00:00:00 | 1267     | 1249        | 698       | 1440      | 1161   |
| 2022-01-02 00:00:00 | 1019     | 1122        | 1227      | 1141      | 1517   |
| 2022-01-03 00:00:00 | 1529     | 1177        | 647       | 1369      | 968    |
| 2022-01-04 00:00:00 | 963      | 1027        | 706       | 1056      | 1537   |
| 2022-01-05 00:00:00 | 590      | 1312        | 1060      | 1089      | 1548   |
| ...                 | ...      | ...         | ...       | ...       | ...    |
| 2023-05-08 00:00:00 | 1035     | 903         | 843       | 1236      | 1021   |
| 2023-05-09 00:00:00 | 1028     | 1145        | 1397      | 1036      | 943    |
| 2023-05-10 00:00:00 | 1349     | 958         | 1254      | 759       | 1053   |
| 2023-05-11 00:00:00 | 200      | 362         | 266       | 232       | 220    |
| All                 | 544593   | 540577      | 549462    | 557665    | 548790 |

| Category            | All     |
|---------------------|---------|
| Date                |         |
| 2022-01-01 00:00:00 | 5815    |
| 2022-01-02 00:00:00 | 6026    |
| 2022-01-03 00:00:00 | 5690    |
| 2022-01-04 00:00:00 | 5289    |
| 2022-01-05 00:00:00 | 5599    |
| ...                 | ...     |
| 2023-05-08 00:00:00 | 5038    |
| 2023-05-09 00:00:00 | 5549    |
| 2023-05-10 00:00:00 | 5373    |
| 2023-05-11 00:00:00 | 1280    |
| All                 | 2741087 |

[497 rows x 6 columns]

```
[447]: pivot = pd.pivot_table(data, values='Competitor Pricing', index='Product ID',
    ↪columns='Date', aggfunc='sum', margins=True, fill_value=0)
print(pivot)
```

| Date       | 2022-01-01 00:00:00 | 2022-01-02 00:00:00 | 2022-01-03 00:00:00 \ |
|------------|---------------------|---------------------|-----------------------|
| Product ID |                     |                     |                       |
| P000       | 0                   | 0                   | 0                     |
| P0001      | 214                 | 303                 | 309                   |
| P0002      | 243                 | 324                 | 275                   |
| P0003      | 244                 | 385                 | 260                   |
| P0004      | 302                 | 360                 | 245                   |
| P0005      | 425                 | 281                 | 273                   |
| P0006      | 338                 | 310                 | 258                   |

|       |      |      |      |
|-------|------|------|------|
| P0007 | 336  | 168  | 326  |
| P0008 | 351  | 277  | 311  |
| P0009 | 320  | 300  | 256  |
| P0010 | 296  | 201  | 324  |
| P0011 | 264  | 313  | 291  |
| P0012 | 254  | 249  | 332  |
| P0013 | 332  | 299  | 276  |
| P0014 | 364  | 280  | 160  |
| P0015 | 250  | 348  | 323  |
| P0016 | 277  | 296  | 326  |
| P0017 | 183  | 271  | 281  |
| P0018 | 241  | 313  | 281  |
| P0019 | 348  | 356  | 318  |
| P0020 | 233  | 392  | 265  |
| All   | 5815 | 6026 | 5690 |

| Date       | 2022-01-04 00:00:00 | 2022-01-05 00:00:00 | 2022-01-06 00:00:00 | \ |
|------------|---------------------|---------------------|---------------------|---|
| Product ID |                     |                     |                     |   |
| P000       | 0                   | 0                   | 0                   |   |
| P0001      | 249                 | 276                 | 258                 |   |
| P0002      | 307                 | 294                 | 252                 |   |
| P0003      | 163                 | 298                 | 249                 |   |
| P0004      | 365                 | 356                 | 314                 |   |
| P0005      | 244                 | 321                 | 260                 |   |
| P0006      | 187                 | 301                 | 281                 |   |
| P0007      | 270                 | 352                 | 282                 |   |
| P0008      | 269                 | 197                 | 218                 |   |
| P0009      | 228                 | 353                 | 251                 |   |
| P0010      | 320                 | 189                 | 309                 |   |
| P0011      | 229                 | 265                 | 311                 |   |
| P0012      | 379                 | 344                 | 297                 |   |
| P0013      | 145                 | 314                 | 312                 |   |
| P0014      | 418                 | 213                 | 269                 |   |
| P0015      | 311                 | 316                 | 268                 |   |
| P0016      | 215                 | 248                 | 350                 |   |
| P0017      | 238                 | 223                 | 200                 |   |
| P0018      | 224                 | 301                 | 319                 |   |
| P0019      | 249                 | 255                 | 279                 |   |
| P0020      | 279                 | 183                 | 395                 |   |
| All        | 5289                | 5599                | 5674                |   |

| Date       | 2022-01-07 00:00:00 | 2022-01-08 00:00:00 | 2022-01-09 00:00:00 | \ |
|------------|---------------------|---------------------|---------------------|---|
| Product ID |                     |                     |                     |   |
| P000       | 0                   | 0                   | 0                   |   |
| P0001      | 341                 | 264                 | 383                 |   |
| P0002      | 389                 | 372                 | 416                 |   |
| P0003      | 286                 | 243                 | 250                 |   |
| P0004      | 277                 | 261                 | 244                 |   |

|       |      |      |      |
|-------|------|------|------|
| P0005 | 300  | 147  | 427  |
| P0006 | 248  | 265  | 344  |
| P0007 | 252  | 262  | 280  |
| P0008 | 271  | 213  | 281  |
| P0009 | 216  | 265  | 361  |
| P0010 | 311  | 339  | 324  |
| P0011 | 299  | 279  | 311  |
| P0012 | 308  | 265  | 294  |
| P0013 | 228  | 371  | 174  |
| P0014 | 297  | 279  | 267  |
| P0015 | 331  | 303  | 235  |
| P0016 | 265  | 183  | 257  |
| P0017 | 305  | 297  | 290  |
| P0018 | 325  | 381  | 230  |
| P0019 | 269  | 126  | 208  |
| P0020 | 339  | 222  | 265  |
| All   | 5857 | 5337 | 5841 |

|            |                     |     |                     |   |
|------------|---------------------|-----|---------------------|---|
| Date       | 2022-01-10 00:00:00 | ... | 2023-05-03 00:00:00 | \ |
| Product ID |                     | ... |                     |   |
| P000       | 0                   | ... | 0                   |   |
| P0001      | 133                 | ... | 239                 |   |
| P0002      | 269                 | ... | 218                 |   |
| P0003      | 284                 | ... | 355                 |   |
| P0004      | 322                 | ... | 256                 |   |
| P0005      | 182                 | ... | 297                 |   |
| P0006      | 275                 | ... | 244                 |   |
| P0007      | 309                 | ... | 295                 |   |
| P0008      | 331                 | ... | 350                 |   |
| P0009      | 279                 | ... | 301                 |   |
| P0010      | 273                 | ... | 323                 |   |
| P0011      | 282                 | ... | 222                 |   |
| P0012      | 246                 | ... | 328                 |   |
| P0013      | 287                 | ... | 302                 |   |
| P0014      | 290                 | ... | 323                 |   |
| P0015      | 221                 | ... | 328                 |   |
| P0016      | 326                 | ... | 277                 |   |
| P0017      | 282                 | ... | 315                 |   |
| P0018      | 259                 | ... | 247                 |   |
| P0019      | 364                 | ... | 224                 |   |
| P0020      | 262                 | ... | 231                 |   |
| All        | 5476                | ... | 5675                |   |

|            |                     |                     |                     |   |
|------------|---------------------|---------------------|---------------------|---|
| Date       | 2023-05-04 00:00:00 | 2023-05-05 00:00:00 | 2023-05-06 00:00:00 | \ |
| Product ID |                     |                     |                     |   |
| P000       | 0                   | 0                   | 0                   |   |
| P0001      | 307                 | 307                 | 244                 |   |
| P0002      | 348                 | 315                 | 288                 |   |

|       |      |      |      |
|-------|------|------|------|
| P0003 | 149  | 248  | 352  |
| P0004 | 312  | 227  | 314  |
| P0005 | 273  | 177  | 301  |
| P0006 | 248  | 297  | 354  |
| P0007 | 303  | 226  | 303  |
| P0008 | 279  | 309  | 299  |
| P0009 | 295  | 180  | 231  |
| P0010 | 185  | 314  | 260  |
| P0011 | 319  | 296  | 301  |
| P0012 | 268  | 176  | 285  |
| P0013 | 232  | 285  | 375  |
| P0014 | 336  | 350  | 328  |
| P0015 | 231  | 222  | 330  |
| P0016 | 233  | 302  | 390  |
| P0017 | 230  | 324  | 185  |
| P0018 | 259  | 276  | 325  |
| P0019 | 291  | 284  | 210  |
| P0020 | 289  | 193  | 341  |
| All   | 5387 | 5308 | 6016 |

| Date       | 2023-05-07 00:00:00 | 2023-05-08 00:00:00 | 2023-05-09 00:00:00 | \ |
|------------|---------------------|---------------------|---------------------|---|
| Product ID |                     |                     |                     |   |
| P000       | 0                   | 0                   | 0                   |   |
| P0001      | 325                 | 312                 | 267                 |   |
| P0002      | 324                 | 336                 | 360                 |   |
| P0003      | 218                 | 319                 | 223                 |   |
| P0004      | 144                 | 257                 | 329                 |   |
| P0005      | 279                 | 151                 | 201                 |   |
| P0006      | 304                 | 199                 | 344                 |   |
| P0007      | 344                 | 291                 | 274                 |   |
| P0008      | 255                 | 243                 | 312                 |   |
| P0009      | 242                 | 173                 | 281                 |   |
| P0010      | 341                 | 264                 | 365                 |   |
| P0011      | 262                 | 247                 | 280                 |   |
| P0012      | 306                 | 201                 | 170                 |   |
| P0013      | 297                 | 307                 | 263                 |   |
| P0014      | 326                 | 252                 | 266                 |   |
| P0015      | 245                 | 279                 | 253                 |   |
| P0016      | 374                 | 129                 | 240                 |   |
| P0017      | 202                 | 238                 | 222                 |   |
| P0018      | 201                 | 278                 | 316                 |   |
| P0019      | 365                 | 227                 | 280                 |   |
| P0020      | 222                 | 335                 | 303                 |   |
| All        | 5576                | 5038                | 5549                |   |

| Date       | 2023-05-10 00:00:00 | 2023-05-11 00:00:00 | All |
|------------|---------------------|---------------------|-----|
| Product ID |                     |                     |     |
| P000       | 0                   | 103                 | 103 |

|       |      |      |         |
|-------|------|------|---------|
| P0001 | 237  | 62   | 136659  |
| P0002 | 261  | 72   | 137723  |
| P0003 | 182  | 70   | 135735  |
| P0004 | 402  | 167  | 138379  |
| P0005 | 210  | 18   | 137014  |
| P0006 | 199  | 42   | 137807  |
| P0007 | 381  | 52   | 135569  |
| P0008 | 254  | 77   | 137485  |
| P0009 | 221  | 14   | 136524  |
| P0010 | 152  | 77   | 138245  |
| P0011 | 270  | 71   | 140837  |
| P0012 | 378  | 82   | 136385  |
| P0013 | 243  | 30   | 136656  |
| P0014 | 313  | 86   | 138258  |
| P0015 | 294  | 81   | 135892  |
| P0016 | 235  | 16   | 135522  |
| P0017 | 255  | 33   | 136544  |
| P0018 | 315  | 11   | 136071  |
| P0019 | 346  | 27   | 136413  |
| P0020 | 225  | 89   | 137266  |
| All   | 5373 | 1280 | 2741087 |

[22 rows x 497 columns]

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### 25.0.6 Conclusion

The retail\_store\_inventory dataset analysis revealed actionable insights for improving inventory management, enhancing profitability, and tailoring strategies to regional demands. By leveraging these insights, the retail store can reduce costs, increase revenue, and better serve customer needs.

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